Enhancing Building Sustainability through Data Analytics: A Study on Energy Efficiency and Emissions Reduction in Catalunya

Malihe Alavi **28 Feb 2025**

1- Introduction

The transition to net-zero buildings is a critical step toward achieving global sustainability goals, particularly in the European Union, where the built environment accounts for a significant portion of energy consumption and greenhouse gas emissions. Improving building energy efficiency not only reduces environmental impact but also lowers operational costs and enhances occupant comfort. With advancements in data analytics and machine learning, it is now possible to analyze building energy performance at an unprecedented scale, identify inefficiencies, and propose targeted interventions. This effort is essential for meeting regulatory standards, combating climate change, and driving innovation in sustainable construction practices.

In this work, I picked the Catalunya dataset with 1,336,925 records and 69 features. I applied a comprehensive EDA and feature engineering on it, however, in some cases where I felt there wasn't enough data for a specific question, I used the Ireland dataset alternatively, which contains 1,264,371 records and 214 features, to address that question. In the last, I trained various ensemble models (RandomForest, CatBoost, XGBoost) on the Catalunya data and used their aggregation to predict the amount of energy consumption (KW/m2) and CO2 emission of each Catalan houses.

The outline of this report is structured as follows:

- **Section 2** will look at the data more closely and explore it briefly
- **Section 3** will delve into the Exploratory Data Analysis (EDA) conducted on the dataset.
- **Section 4** will detail the Feature Engineering methods applied to the data.
- **Section 5** will focus on the model, Bonus Question, and will discuss the interpretability of the model using tools like SHAP and LIME.
- **Section 6** will present the Appendix section.

2- Data Overview

Catalunya dataset contains1,336,925 records and 69 features. Features can be divided into six categories:

- Geographic information: Province, County, Street, latitude and longitude, address, Municipality, …
- Energy consumption/ CO2 emission: these features exist in both categorical rating values and real continues values
    - Rating: Non-renewable primary energy rating letter (AG), CO2 emission rating letter (AG), Qualification of CO2 emissions associated with the heating / refrigeration / lightening/ Domestic Hot Water service (AG), Qualification of non-renewable primary energy consumption by the heating / refrigeration / lightening/ Domestic Hot Water service (AG), …
    - Real Continuous: Non-renewable primary energy value [kWh / m2 · year], Value of CO2 emissions [kg CO2/m2 any] , Energy at the point of consumption equivalent to energy consumption [kWh / m2 · year] , CO2 emissions associated with the heating / refrigeration / lightening/ Domestic Hot Water service [kg CO2 / m2 · year] , non-renewable primary energy consumption by the heating / refrigeration / lightening/ Domestic Hot Water service [kWh / m2 · year]
- Financial Cost: Approximate annual cost of energy per home
- House Features: It has an electric vehicle charging point (Yes / No), It has solar thermal installation (Yes / no), It has photovoltaic installation (Yes / no), Average window transmittance [W / m2 · K], It has biomass installation (Yes / no), …
- Renovation Measures: Rehabilitation actions, Energy rehabilitation has been carried out [Yes / no]
- Date: year of construction , Date of registration

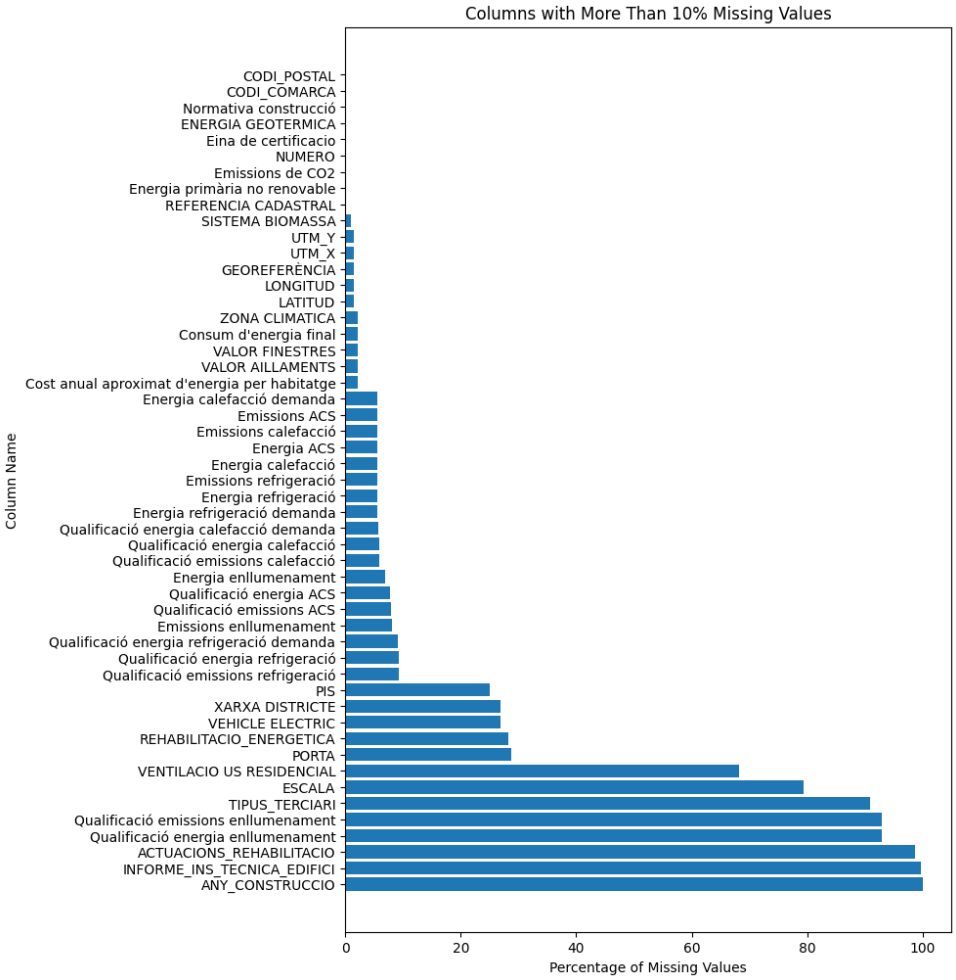
2-1 Missing values

most columns of Catalunya dataset have few missing values only a few columns are sparse. Figure 1 illustrates the percentage of missing values of each feature. 6 features have more than 90% missing values, which are listed below. The two most important columns among these are ACTUACIONS_REHABILITACIO (Rehabilitation actions) and ANY_CONSTRUCTO (Year of construction)

Table 1- list of columns with more than 60% missing values

	column_name	number_of_missing_values	percentage_of_missing_values
ESCALA	ESCALA	1061709	79.414253
ANY_CONSTRUCCIO	ANY_CONSTRUCCIO	1336602	99.975840
INFORME_INS_TECNICA_EDIFICI	INFORME_INS_TECNICA_EDIFICI	1331865	99.621520
TIPUS_TERCIARI	TIPUS_TERCIARI	1214326	90.829777
Qualificació emissions enllumenament	Qualificació emissions enllumenament	1241627	92.871851
Qualificació energia enllumenament	Qualificació energia enllumenament	1241677	92.875591
VENTILACIO US RESIDENCIAL	VENTILACIO US RESIDENCIAL	912631	68.263440
ACTUACIONS_REHABILITACIO	ACTUACIONS_REHABILITACIO	1319813	98.720048

Figure 1- percentage of missing values



More specific detail about the number of columns with particular missing values is listed below. We can see that only 8 columns have more than 50% missing values and 56 columns have less than 10% missing values. 5 columns also have between 30% to 50 percent missing values.

Figure 2 number of columns with particular missing values

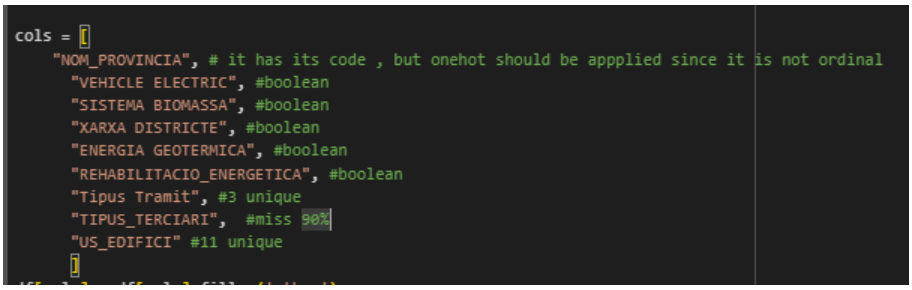
```
number of fully poplulated columns: 18
number of columns with missing values less than 10% :  56
number of columns with missing values less than 20% :  56
number of columns with missing values less than 30% :  61
number of columns with missing values less than 40% :  61
number of columns with missing values less than 50% :  61
number of columns with missing values more than 50% :  8
number of columns with missing values more than 80% :  6
number of columns with missing values more than 90% :  6
```

2-2 Data Type

40 features have string datatype and 29 of them were either float64 or int64. Among these 40 string features, 13 of them were ordinal features (except climate zone , the other 12 features are energy-rating features that their name starts with Qualificació). These features have values from A-G which A demonstrate the best quality and G is the worst quality. All these ordinal features were encoded with label encoding method.

9 of these 40 features were non-ordinal features with low cardinality so I encoded them with one-hot encoding method. These features are:

Figure 3 Non-ordinal features with low cardinality



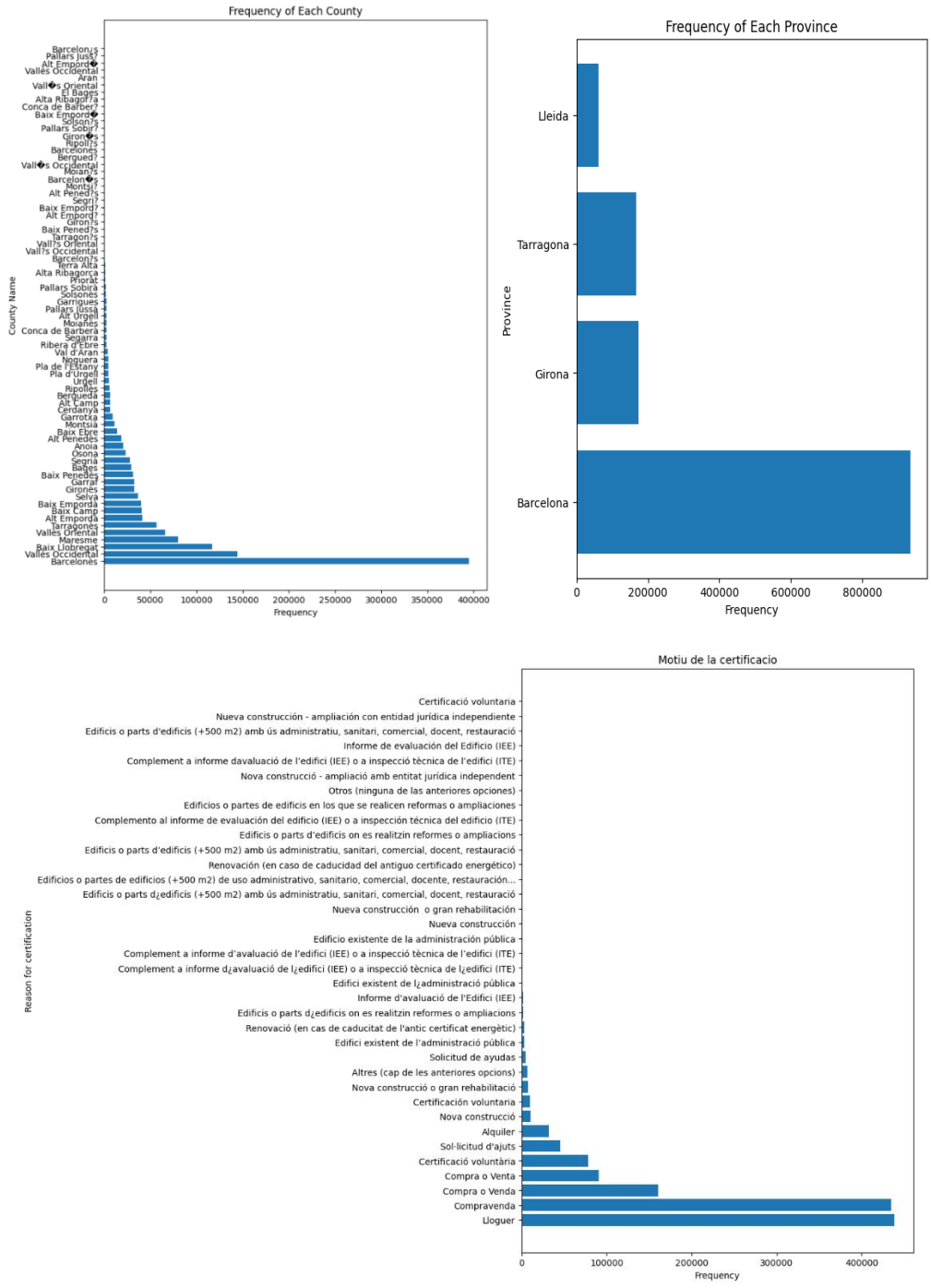
4 of these string features were non-ordinal but with high cardinality These feature later were encoded using frequency encoding techniques. The last two features in the below image had integer datatype but they were categorical data with a huge gap between their values' frequency. These two also were encoded by frequency encoding.

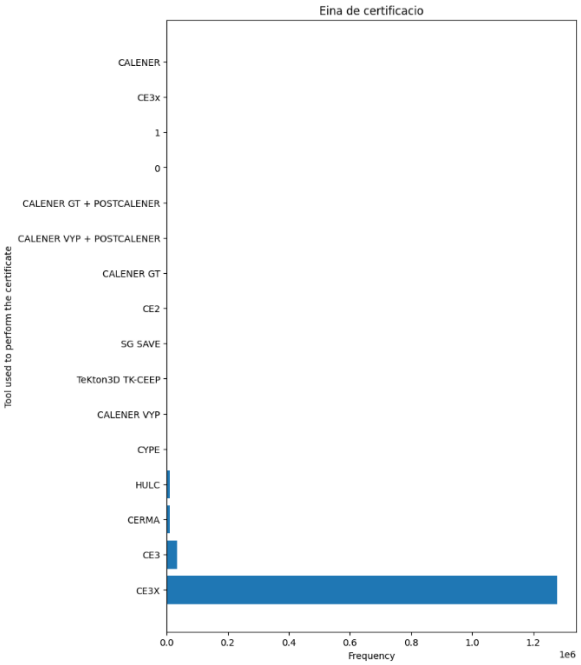
Figure 4 Non-ordinal features with high cardinality

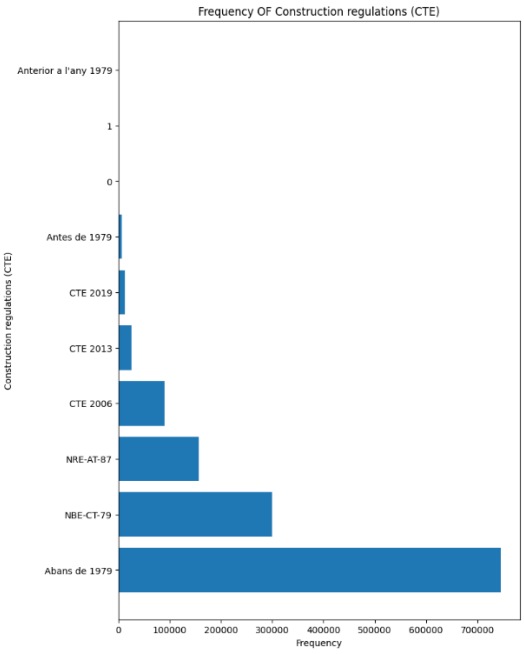


Figure 5 shows the frequency of values of these features. These figures indicate a strong geographical concentration, with Barcelona being the most represented province, far outnumbering others like Girona, Tarragona, and Lleida. In terms of energy certification tools, "CE3X" is the most commonly used, followed by "CE3" and other tools. The construction regulations also show a clear trend, with pre-1979 regulations being the most frequent, followed by more recent standards such as the NBE-CT-79 and others.

Figure 5 Frequency bar plot of features







3- EDA

For the first two questions in the EDA part, I focused on ACTUACIONS_REHABILITACIO (Rehabilitation actions) feature. This feature as I mentioned before, has 98.7 % missing values, where comparing it with REHABILITACIO_ENERGETICA feature which has boolean data type and implies whether houses have applied renovation actions or not, indicates that these amounts of missing values imply 98.7% of houses haven't applied renovation actions.

```
[ ]  print('number of houses which have done renovation: ', len(df[df['REHABILITACIO_ENERGETICA']=='SI']))
     print('percentage of houses which have done renovation: ', len(df[df['REHABILITACIO_ENERGETICA']=='SI'])/len(df['REHABILITACIO_ENERGETICA']))

     number of houses which have done renovation:  18326
     percentage of houses which have done renovation:  0.013707575219253136
```

ACTUACIONS_REHABILITACIO

17,112 records in this feature have values that means this amount of houses have already perform renovations. This feature has unstructured messy text data in Catalan language, some records have multiple renovations separated with || which need to be split and organized. Here is a raw record of ACTUACIONS_REHABILITACIO.

Aïllament en façanes i/o coberta. || Renovació de finestres i/o proteccions solars. || Millora de les instal·lacions (calefacció, climatització, ACS, enllumenat,...)

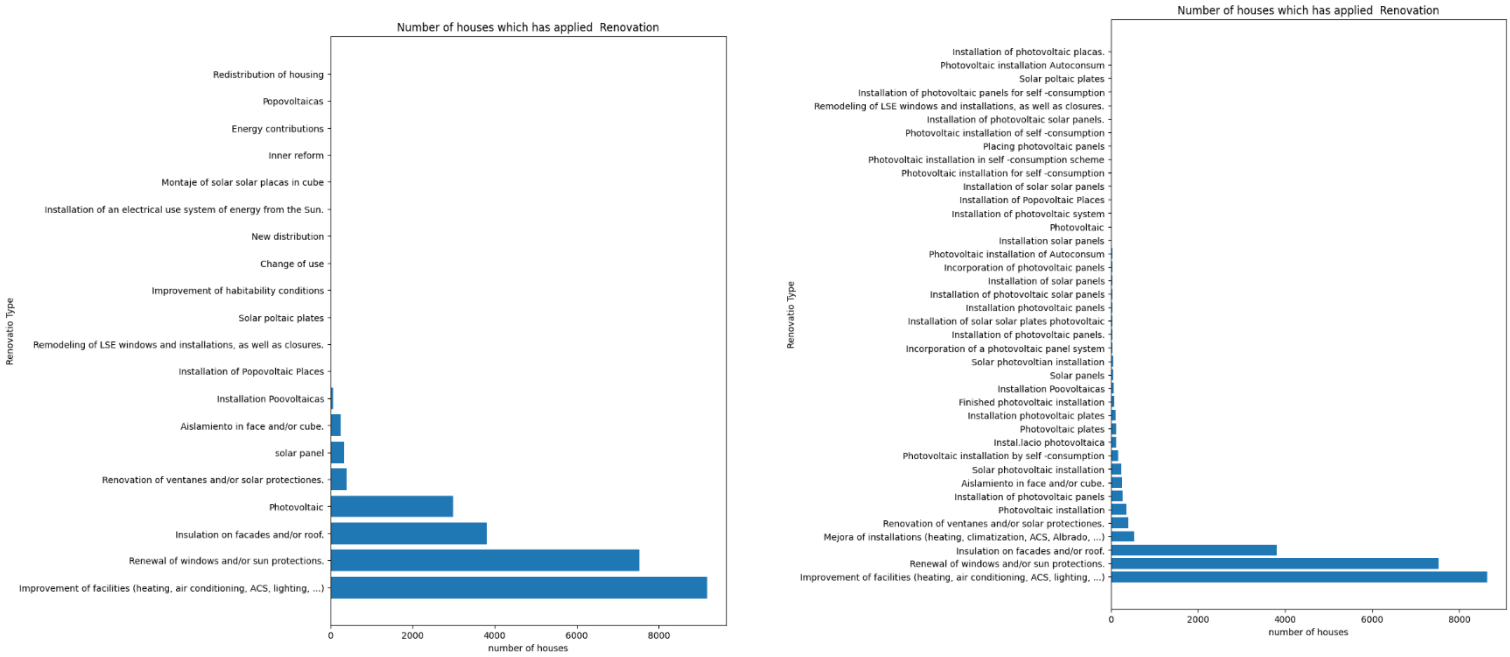
The following preprocessing steps were applied on this feature:

1- Removing redundant characters like [], {} , || , ...
2- Translation of all Catalan text into English text using *googletrans* library in python
3- Splitting the records with multiple renovations into multiple records.

4- Grouping and aggregating records with similar context but different wording/structure (there were numerous records of Photovoltaic Installation which had different wording, I tried to group them as much as possible but some records with very different shape still remained)
5- Counting each group's frequency and pick the top most frequent groups: since after grouping them there were still plenty of categories of renovation (about 1114 unique categories) which most of them was very infrequent and rare, I set a small threshold (0.0005) and only picked those which had a frequency of at least 0.05% of the total dataset. I grouped the rest of them in 'other' category. 11 renovation categories were chosen for further analysis

Figure 6 Frequency plot of renovation measures.

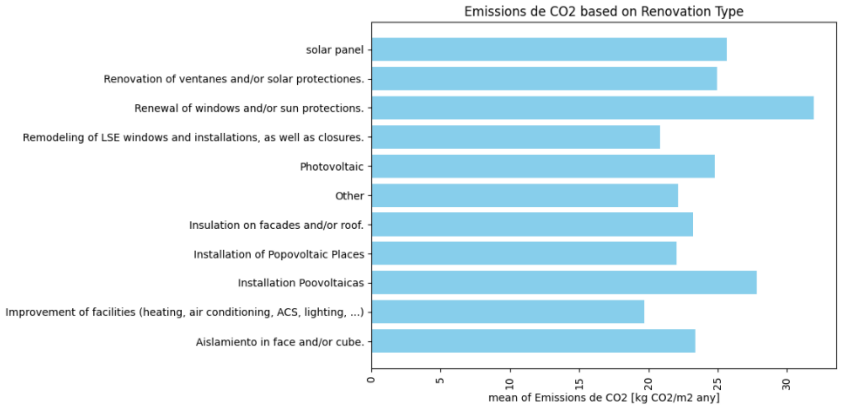
The right figure is before grouping, the left figure is after grouping



3-1 Identify the most cost-effective renovation measures in terms of energy reduction (kWh/m^2 saved per cost unit).

There were multiple features in the dataset that demonstrate energy consumption and CO2 emissions. Each of these was investigated separately based on the processed categorized renovation measures to see which measure is more effective.

Figure 7 the amount of **CO2 emission** based on each renovation measure

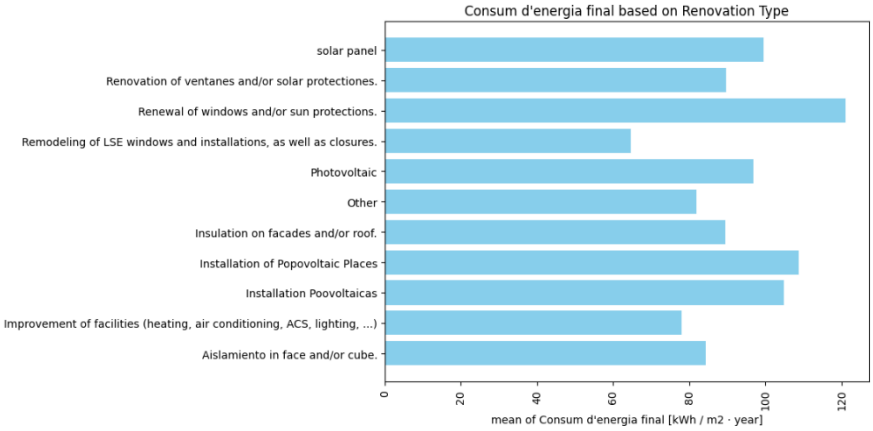


	grouped_translated	mean	median	std	count
0	Aislamiento in face and/or cube.	23.364585	19.240	17.965359	253
1	Improvement of facilities (heating, air condit...	19.703294	15.970	14.758063	9173
2	Installation Poovoltaicas	27.840645	25.930	18.305020	62
3	Installation of Popovoltaic Places	22.004545	18.140	17.528732	22
4	Insulation on facades and/or roof.	23.239932	18.430	17.984470	3810
5	Other	22.143931	18.745	16.114891	1328
6	Photovoltaic	24.781618	22.030	18.041134	2992
7	Remodeling of LSE windows and installations, a...	20.853077	21.900	2.662663	13
8	Renewal of windows and/or sun protections.	31.957367	28.150	23.613232	7521
9	Renovation of ventanes and/or solar protectiones.	24.958589	22.240	16.313664	397
10	solar panel	25.663988	24.000	15.619087	341

Figure 7, demonstrates the amount of CO2 emission based on each type of renovation. The right table shows mean, median, std of CO2 emissions [kg CO2/m2 any] and the number of houses that have applied that type or renovation. The left figure visualizes the mean of CO2 emission of houses placing into that group of renovation. Looking at these two figures, *Improvement of facilities (heating, air condition, ...)* has worked better than the other renovations in both terms of mean and median to reduce the emission of CO2. This measure is the most popular one among Catalan houses where more than 9000 houses have applied it.

Figure 8 Value of **Energy at the point of consumption equivalent to energy consumption**

based on each renovation measure

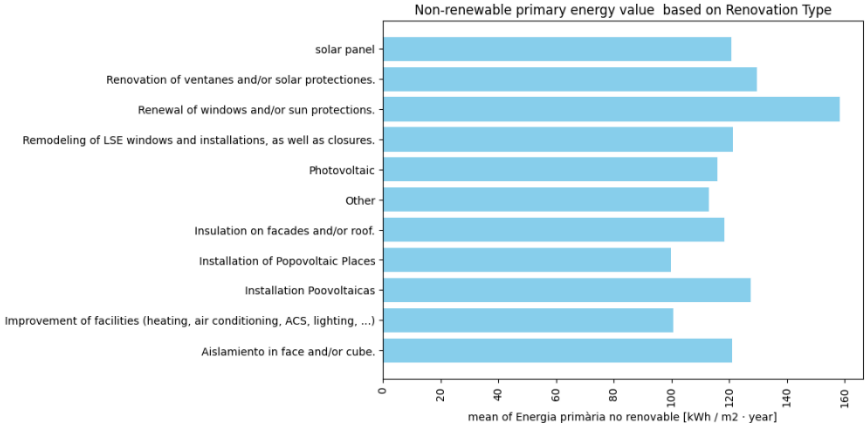


	grouped_translated	Consum d'energia final			
		mean	median	std	count
0	Aislamiento in face and/or cube.	84.374672	65.655	73.079985	244
1	Improvement of facilities (heating, air condit...	77.972948	60.160	68.250390	9105
2	Installation Poovoltaicas	104.789677	96.960	65.774400	62
3	Installation of Popovoltaic Places	108.810909	80.940	89.688595	22
4	Insulation on facades and/or roof.	89.401544	70.340	72.392421	3751
5	Other	81.783171	68.030	64.117750	1315
6	Photovoltaic	96.768191	86.290	71.912700	2991
7	Remodeling of LSE windows and installations, a...	64.556923	66.410	10.241490	13
8	Renewal of windows and/or sun protections.	121.130492	107.320	88.770941	7454
9	Renovation of ventanes and/or solar protectiones.	89.575299	77.560	63.982733	385
10	solar panel	99.484428	91.750	59.085441	341

Figure 8 illustrates the Energy at the point of consumption equivalent to energy consumption (*Consum d'energia final*) associated with each type of renovation. The right table provides the mean, median, standard deviation of energy consumption in a year (KW/m2-year), and the number of houses that have undergone that specific type of renovation. The left figure visualizes the mean energy consumption of houses categorized by renovation type. Observing these two figures, it is evident that the *Remodeling of LSE windows and installations as well as closure*, has been more effective than other renovations in term of mean. However, this renovation is the least frequent with

only 13 houses having applied it. Setting it aside, *Improvement of facilities (heating, air condition, ACS, lightening)* outperforms the other measures in terms of both mean and median.

Figure 9 the value of **Non-renewable primary energy** based on each renovation measure



	grouped_translated	mean	median	std	count
0	Aislamiento in face and/or cube.	121.046482	103.520	85.139733	253
1	Improvement of facilities (heating, air condit...	100.737327	85.350	71.354904	9173
2	Installation Poovoltaicas	127.576774	110.630	80.798859	62
3	Installation of Popovoltaic Places	99.744545	83.655	72.926716	22
4	Insulation on facades and/or roof.	118.405110	97.620	85.441497	3810
5	Other	112.948825	98.130	79.861133	1328
6	Photovoltaic	115.806250	105.590	81.061414	2992
7	Remodeling of LSE windows and installations, a...	121.153077	129.280	16.602821	13
8	Renewal of windows and/or sun protections.	158.430468	144.350	110.754762	7521
9	Renovation of ventanes and/or solar protectiones.	129.609345	117.750	78.624423	397
10	solar panel	120.713255	114.520	66.766339	341

Figure 9 shows the Non-renewable primary energy (*Energia primària no renovable*) consumption in a year (KW/m2-year), associated with each type of renovation. Based on these images, *Installation of Popovoltaic* Places with mean 99.7 KW/m2 and median 83.6 KW/m2 in a year and *Improvement of facilities (heating, air condition, ACS, lightening)* with mean 100.7 KW/m2 and median 85.3 KW/m2 in a year have major roles in reducing Non-renewable primary energy consumption.

Figure 10 Overall Comparison

By comparing Figures 7-10, one can conclude that *Improvement of facilities (heating, air condition, ACS, lightening)* can be considered as the best actions and *Renewal of windows and/or sun protections* is the least effective measure in reducing energy consumption and CO2 emissions. These measures have been applied in nearly 9000 and 7500 houses respectively.

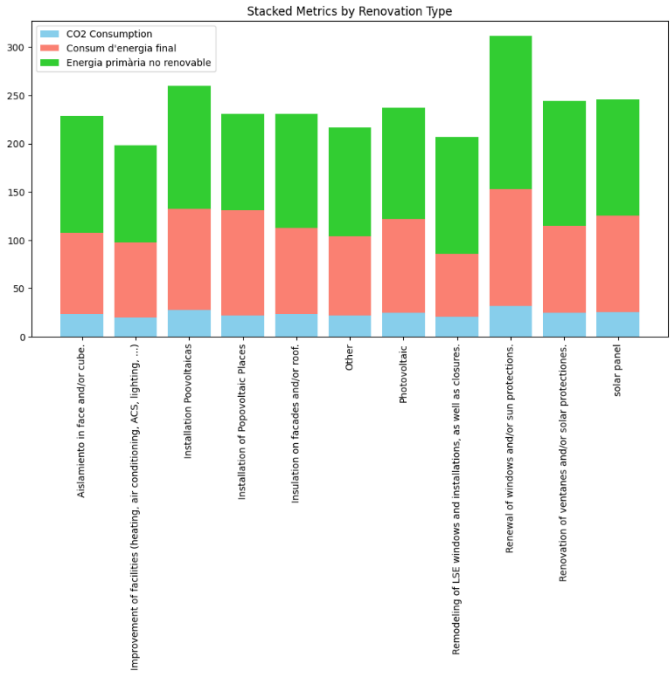


Figure 11 Comparison of Energy Consumption/CO2 Emissions with or without renovation actions

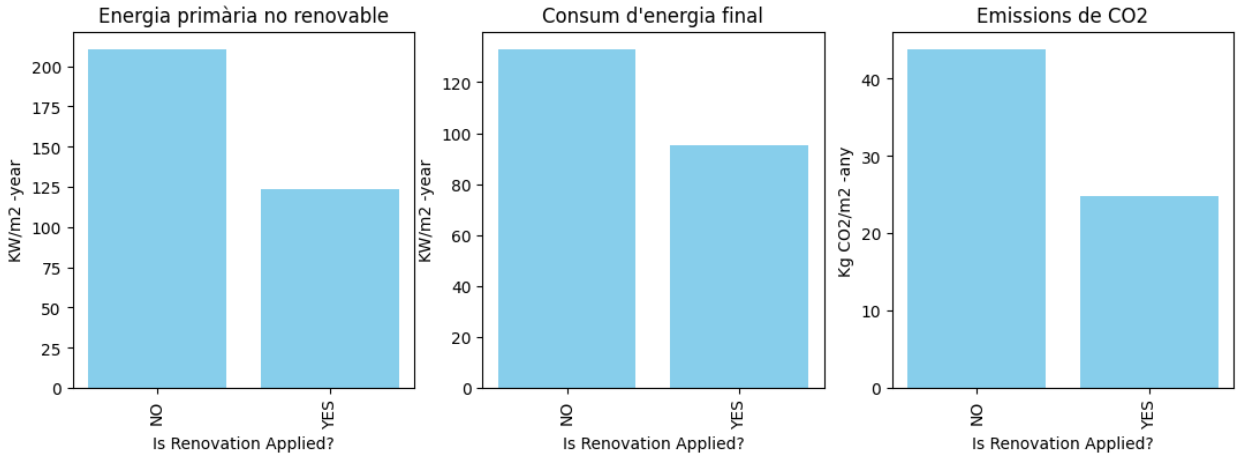
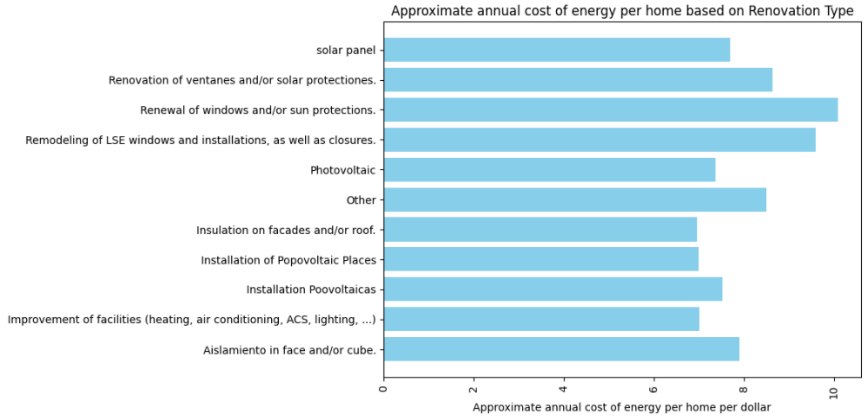


Figure 11 presents a comparison of three key metrics—non-renewable primary energy, final energy consumption, and CO2 emissions—with or without renovation actions. The bars show a clear reduction in all three categories when renovation is applied. Specifically, non-renewable primary energy consumption drops from over 200 units to below 125 units, final energy consumption decreases from around 130 units to just below 100 units, and CO2 emissions are reduced from over 40 units to around 30 units. This indicates that renovation actions lead to significant improvements in energy efficiency and a reduction in environmental impact.

3-2 Determine the most financially efficient renovation measures in terms of energy cost savings per investment dollar.

Feature *Cost anual aproximat d'energia per habitatge* indicates Approximate annual cost of energy per home. In order to determine the most financially efficient renovation measures, I calculated the mean, median and standard deviation of each renovation measure in term of this feature.

Figure 12 the impact of renovation actions on **annual cost of energy per home**



	grouped_translated	mean	median	std	count
0	Aislamiento in face and/or cube.	7.897828	6.725	5.882529	244
1	Improvement of facilities (heating, air condit...	7.019537	5.880	5.412156	9105
2	Installation Poovoltaicas	7.526774	7.000	5.122675	62
3	Installation of Popovoltaic Places	6.996364	5.660	5.022149	22
4	Insulation on facades and/or roof.	6.957329	5.680	5.930049	3751
5	Other	8.499894	6.140	41.617420	1315
6	Photovoltaic	7.363477	6.410	5.925889	2991
7	Remodeling of LSE windows and installations, a...	9.596154	10.490	1.817235	13
8	Renewal of windows and/or sun protections.	10.097678	9.110	7.737821	7454
9	Renovation of ventanes and/or solar protectiones.	8.640519	7.700	5.788433	385
10	solar panel	7.690000	7.190	4.663472	341

Figure 12 illustrates the impact of different renovation actions on annual cost of energy per home. It is evident that Window-focused renovations, such as "Renewal of windows and/or sun protections" and "Remodeling of LSE windows," show substantial energy effects but are among the most expensive options, with costs around 10-11 dollars per home. In contrast, solar panels are more cost-effective at approximately 8 dollars per home, while insulation on facades/roofs offers moderate costs of about 7 dollars. The "Improvement of facilities" is the most prevalent renovation type in Catalunya, applied to a large number of homes. Not only is it effective in reducing energy consumption, but it is also relatively cost-effective, with costs averaging around 7 dollars per home. This combination of efficiency and affordability likely explains why it is the preferred choice for many homeowners in the region.

Figure 13 financial impact of renovation actions

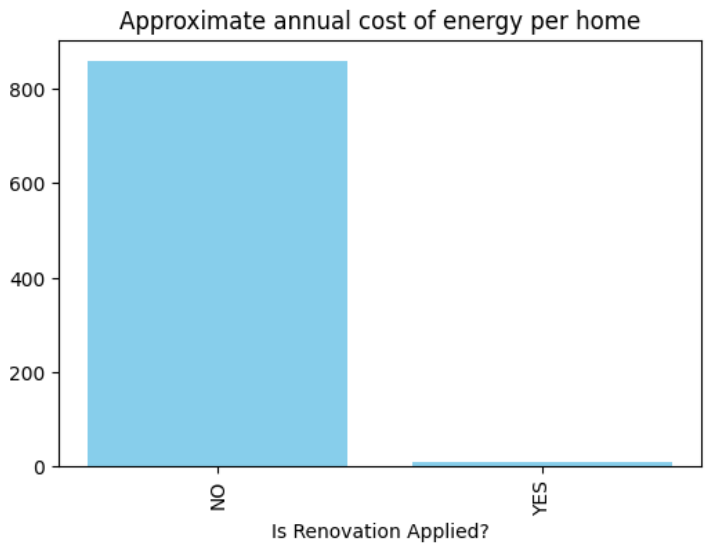


Figure 13 dramatically illustrates the financial impact of renovations, showing approximate annual energy costs per home with and without renovation. Homes without renovations ("NO") have costs around 850 units (likely dollars or euros), while renovated homes ("YES") show drastically reduced costs near zero. This stark contrast demonstrates that renovations, despite their varying effectiveness shown in Figure 12, collectively result in substantial energy cost savings. The magnitude of this difference suggests that virtually any of the renovation categories from the first image would likely produce significant financial benefits for homeowners, though some (particularly window-related renovations) may offer superior performance.

3-3 Analyze which building elements have the greatest impact on energy consumption.

For this part, Catalunya dataset lacked the features that address building elements. It contained only four boolean features which doesn't completely reflect building elements, e.g.

- VEHICLE ELECTRIC: It has an electric vehicle charging point (Yes / No)?
- SISTEMA BIOMASSA: It has biomass installation (Yes / no)?

- XARXA DISTRICTE: It has connection to a district network of generation of heat and / or cold (Yes / no)?
- ENERGIA GEOTERMICA : It has geothermal installation (Yes / no)?

Therefore, I also used the Irland dataset for this part since this dataset has more features related with building elements. Figures 15-19 represents the information of the Irland dataset.

Figure 14 Impact of Catalunya buildings' elements on Energy Consumption

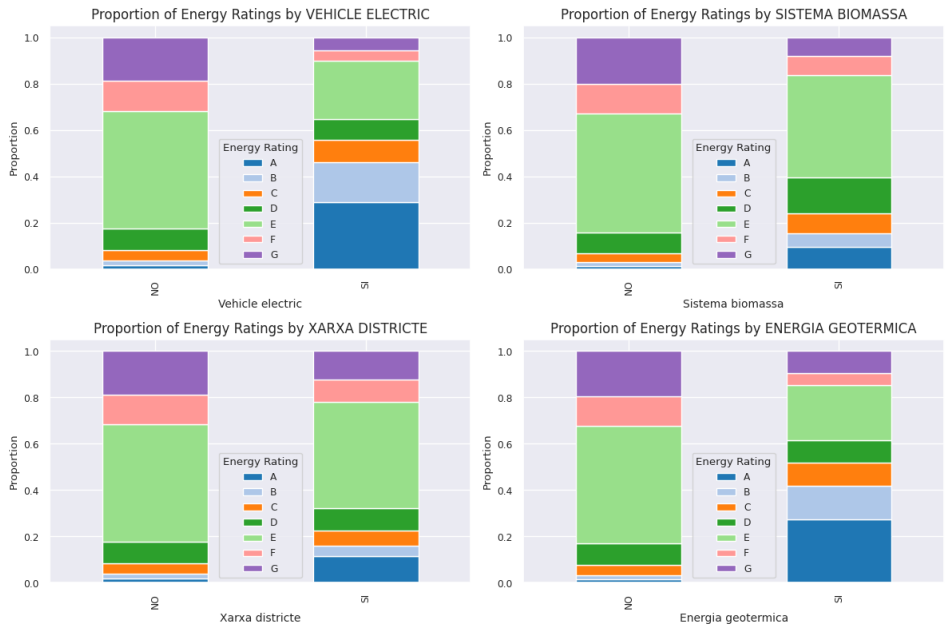


Figure 14 presents a comparative analysis of energy ratings across four distinct elements under two different conditions, denoted as "NO" and "SI." A visual inspection reveals that the distribution of energy ratings varies significantly among the elements and between the two conditions. In general, a shift towards higher energy ratings (e.g., A and B) is observed in the "SI" condition compared to the "NO" condition across all elements, suggesting an improvement in energy performance.

Specifically, the element labeled "Energia Geotermica" exhibits the most pronounced shift, with a majority of instances moving to the highest energy rating under the "SI" condition. Similarly, "Vehicle Electric" also demonstrates a substantial improvement, with a large proportion achieving high energy ratings in the "SI" scenario. The remaining two elements, "Xarxa Districte" and "Sistema Biomassa," also show positive trends, albeit to a lesser extent, with a noticeable migration towards better energy ratings when comparing the "NO" and "SI" states.

Figure 15 The impact of **type of Roof** on Energy

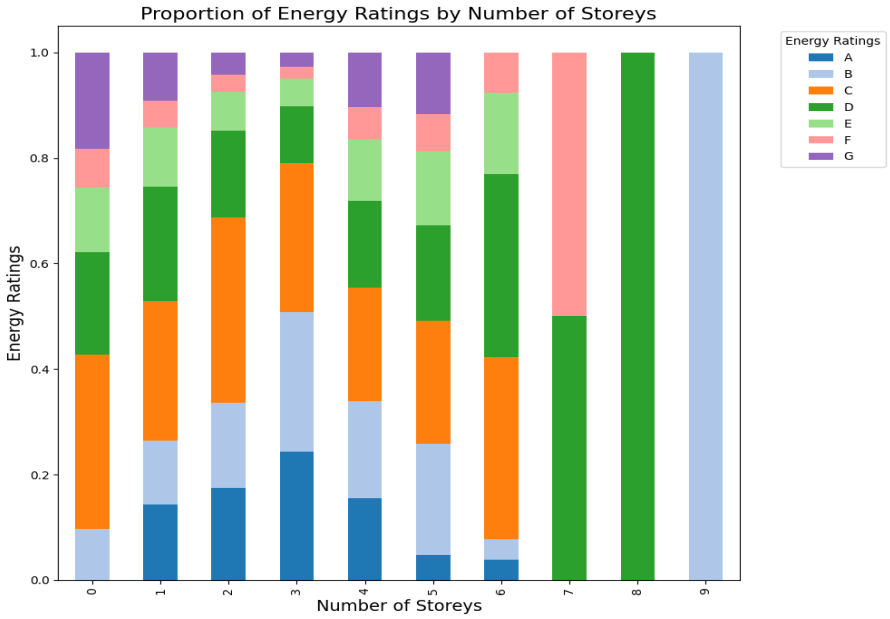


Figure 15 illustrates the proportion of energy ratings relative to the number of storeys in a building. Buildings with zero storeys have a wide range of energy ratings, while buildings with one or two storeys are more likely to have B or C ratings. As the number of storeys increases, the proportion of lower energy ratings (E, F, G) tends to increase, with a noticeable shift towards lower ratings for buildings with 6 or more storeys. Buildings with nine stories are rated B, which probably could be commercial building with high renewable energy technologies, or simply be an outlier.

Figure 16 The impact of **number of building's storeys** on Energy

Figure 16 presents the proportion of energy ratings based on the predominant roof type. Flat roofs and "other" roof types show a diverse mix of energy ratings, while pitch roofs with insulation on the ceiling or rafter tend to have better energy ratings (A, B, C). Buildings with "room in roof insulation on side" also show a mix of ratings. The "select roof type" category is overwhelmingly dominated by the E rating, indicating this roof type is associated with poorer energy performance compared to the others.

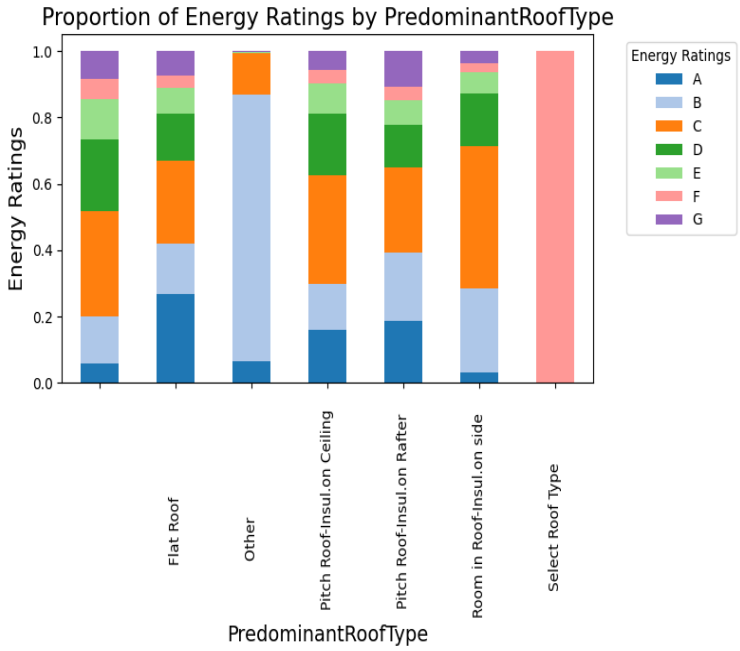


Figure 17 stacked bar charts of different building elements and Energy Rating

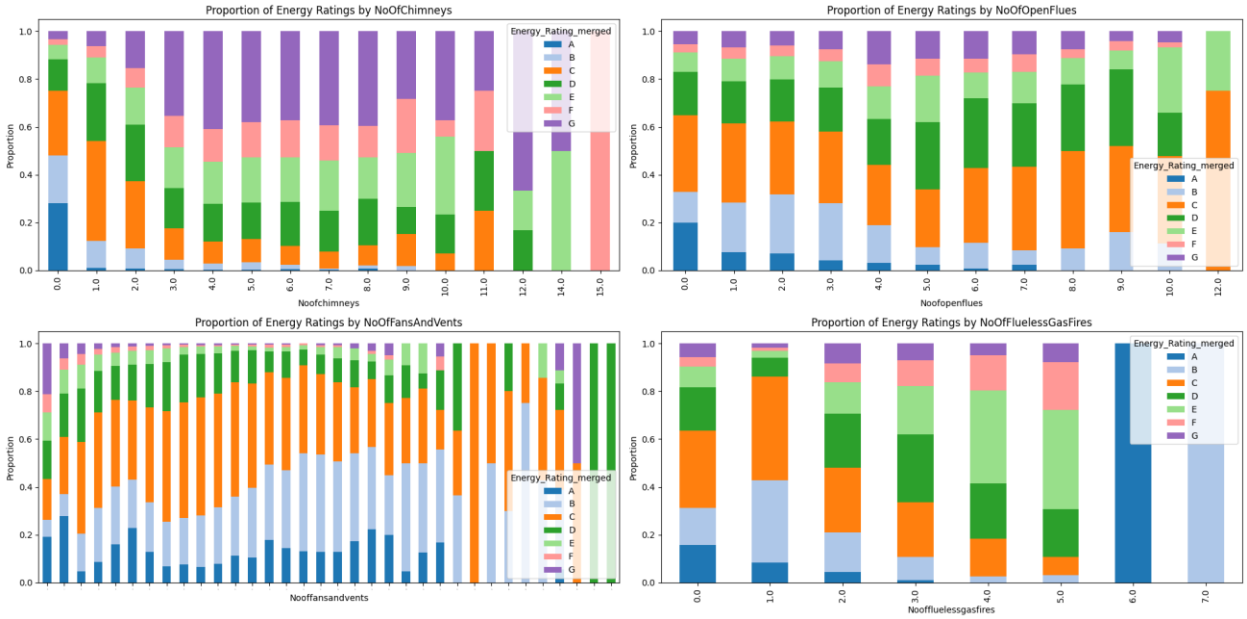


Figure 17 presents a series of stacked bar charts illustrating the relationship between different building elements (number of chimneys, open flues, fans and vents, and flueless gas fires) and the energy ratings of buildings. Based on this image:

- **Number of Chimneys:** The plot illustrates that buildings with very few or a high number of chimneys tend to have lower energy ratings (more F and G ratings). Buildings with a moderate number of chimneys (around 4-8) appear to have a slightly better distribution of energy ratings.
- **Number of Open Flues:** The plot shows a general trend where buildings with fewer open flues tend to have better energy ratings (more A and B ratings). As the number of open flues increases, the proportion of lower energy ratings (E, F, and G) also tends to increase.
- **Number of Fans and Vents:** This plot indicates that buildings with very few fans and vents tend to have higher energy ratings. As the number of fans and vents increases, the energy ratings become more distributed, with a larger proportion of lower ratings.
- **Number of Flueless Gas Fires:** The plot suggests that buildings with no flueless gas fires have the best energy ratings. As the number of flueless gas fires increases, the proportion of lower energy ratings increases significantly.

Figure 19 the financial impact of renovations

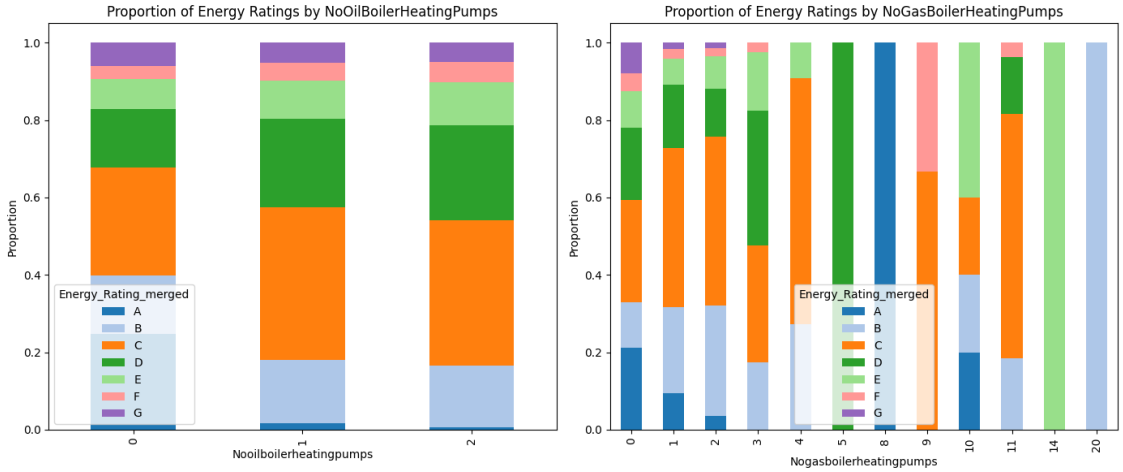


Figure 18 presents two stacked bar charts comparing the distribution of energy ratings in buildings based on the number of "Oil Boiler Heating Pumps"(left) and "Gas Boiler Heating Pumps"(right). Left chart indicates that houses have (0, 1, or 2) Oil Boiler Heating Pumps. With zero pumps, the majority of buildings have a C rating, followed by D. As the number of these pumps increases to 1 and then 2, there's a slight shift: the proportion of B-rated buildings increases noticeably, while the proportion of C-rated buildings decreases a bit, suggesting a potential correlation between having these pumps and achieving a slightly better energy rating.

The right chart shows the distribution of energy ratings based on the number of " Gas Boiler Heating Pumps." Here, we see more variance. Buildings with very few or many of these pumps (0, 1, 11, 14, and 20) tend to have a higher proportion of A and B ratings. In contrast, buildings with an intermediate number of these pumps (around 6 to 10) appear to have a larger proportion of lower energy ratings like C and D. This suggests a more complex relationship where having a certain number of " Gas Boiler Heating Pumps " might not linearly correlate with better energy performance; other factors are likely at play.

4- Feature Engineering

Categorical Data

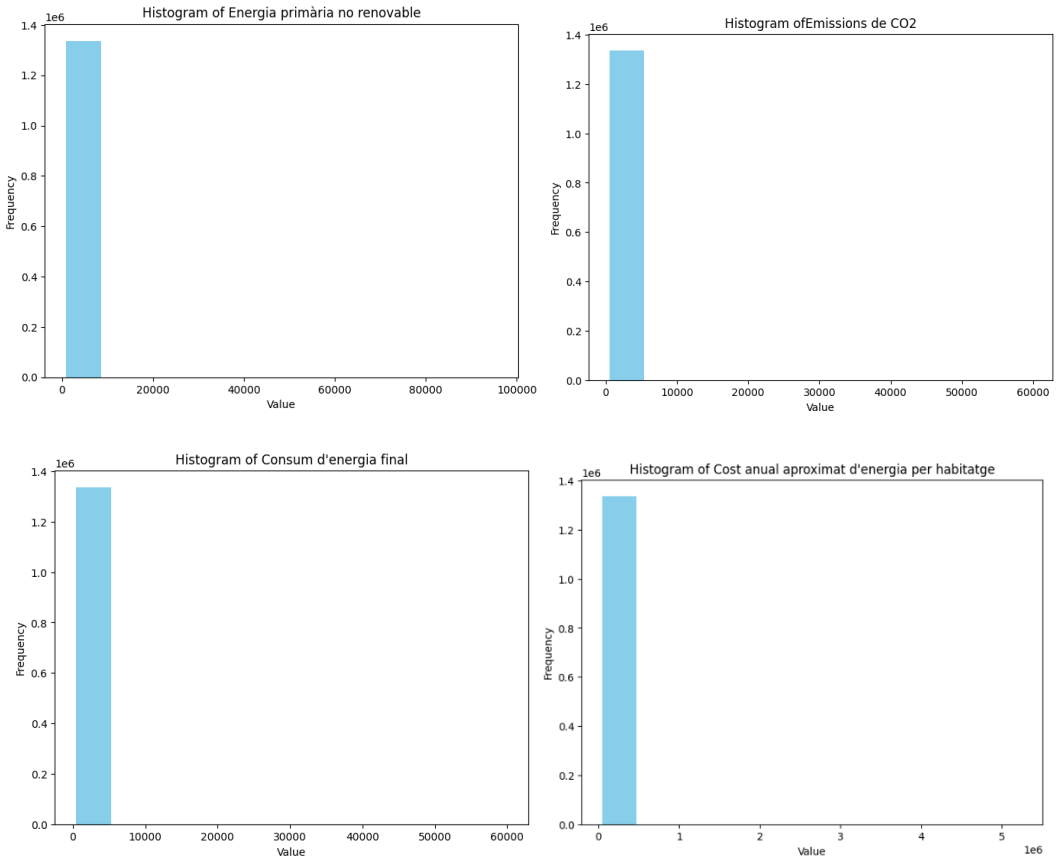
there were around 28 categorical features in this dataset (both string and float/ int type). 13 of them were ordinal which indicates rating of energy consumption/ CO2 emissions and all these features encoded by label encoding to preserve the order, nan values in these features replaced with a very big number i.e. “99999”. 9 of features were encoded by one hot encoding since they were how cardinal and non-ordinal features, nan values in these features were removed after applying one-hot encoder. 6 features with high cardinality were encoded using frequency encoding where the frequency of each record replaced with the real value.

after performing the encoding process, I standardized these features using standard scalar.

Continuous Data

Most of the continuous features in the data show a large concentration of data around the lower values (close to zero) and a long tail extending towards higher values. Where the data is highly skewed. Here are the histogram plot of some raw features before any transformation:

Figure 20 the distribution of raw continuous data



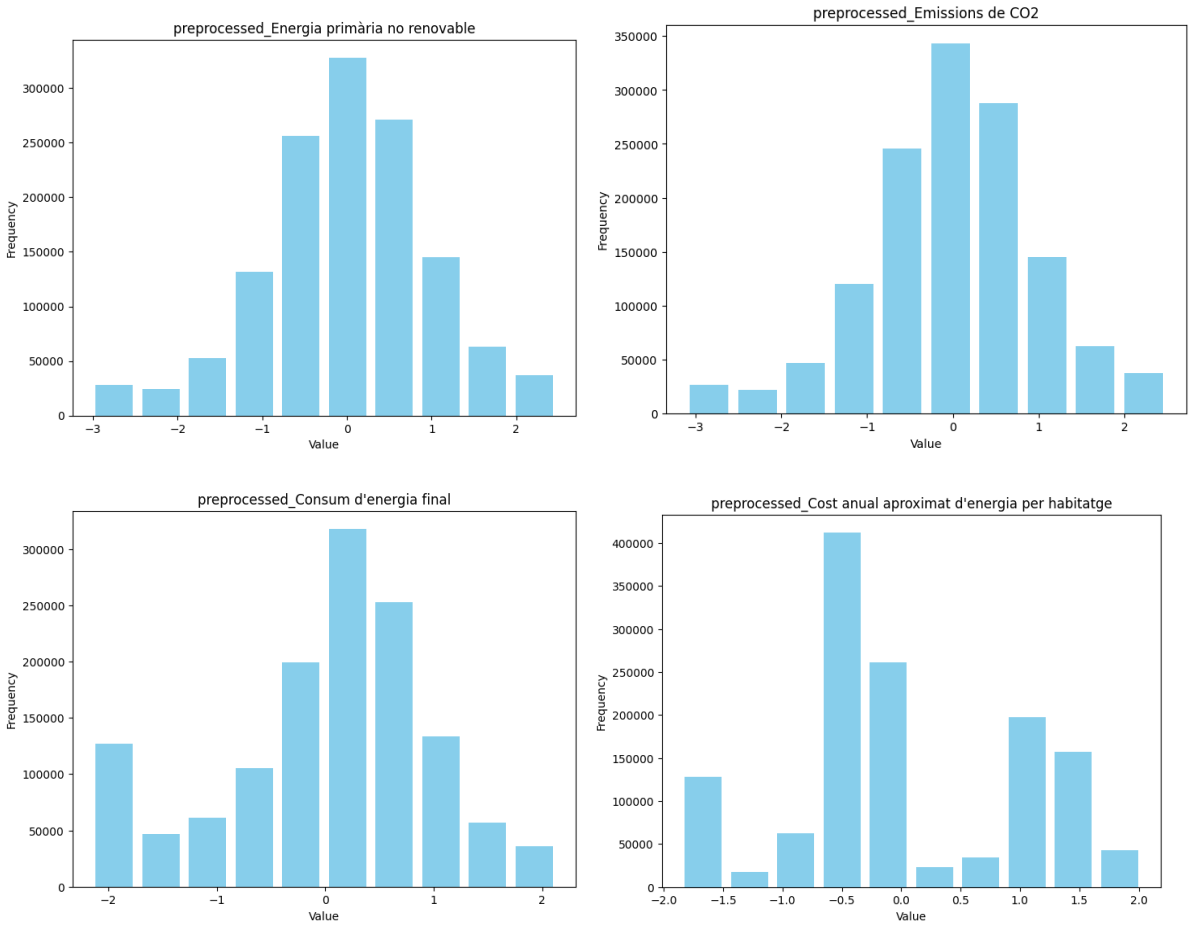
Other features had the same distribution which make it hard to interpret and train a model on them. To deal with this issue, I passed each feature through 4 processing steps to remove the negative values, avoid the outliers, uniform the distribution and standardize the scale of them. Here is the code snippet for doing so:

```
def preprocessing_numerical_continuous_feature(df, feature, ax=None):

  name= 'preprocessed_' +feature
  # 1- replacing all negative values with zero
  df[name] = df[feature].apply(lambda x: x if x >= 0 else 0)
  # 2- boxcox transformation to unifor the data
  df[name], _ = stats.boxcox(df[name] + 1)
  # 3- removing outliers of begger and smaller 99 and 1 percentile
  lower_percentile = np.percentile(df[name], 1)
  upper_percentile = np.percentile(df[name], 99)
  df[name] = np.clip(df[name], lower_percentile, upper_percentile)
  # 4- standardize the scale
  scaler = StandardScaler()
  df[name] = scaler.fit_transform(df[[name]])
```

After applying these steps features became more normalized and ready to be trained. Figure 21 shows some of these features after transformation.

Figure 21 the distribution of preprocessed continuous data



The feature of cost still doesn't follow a uniform distribution but I didn't transform it again to preserve the nature of the data.

New Features

By looking at the data some new potential features were created which some of them turned out to have significant impact on the modeling process. Here are the list of features:

- Age: this feature was created by subtracting the current year (2025) from the year of DATA_ENTRADA (Date of registration).
- Energy Efficiency: for each of the four metrics of heating(calefacció), lightening (enllumenament) , domestic hot water (ACS) and cooling (refrigeració) I created new feature to address the efficiency of these metrics by dividing their energy consumption to their CO2 emissions. For example here is how heating-efficiency feature was made: df['heating_efficiency'] = df['Energia calefacció'] / df['Emissions calefacció']

- Total Energy : this feature was created by adding up the four above energy consumption's metrics. i.e. heating(calefacció), lightening (enllumenament) , domestic hot water (ACS) and cooling (refrigeració)
- Total Emission : this feature was created by adding up the four above CO2 emissions' metrics.

These features then transformed and normalized to be prepared for the model.

5- Model

For modeling section, I used three ensemble model XGboost, Catboost, Random Forest and one stacking model which aggregates these three trained models using a simple linear regression for more robustness. The reason of choosing these three ensembles is due to their robust performance, efficiency, and ability to handle complex data. XGboost and Catboost also execute too fast which make them appropriate for big scale of data.

I picked two target to be predicted: CO2 emission (*Emissions de CO2*) and final Energy Consumption (*Consum d'energia final*). I'll explain the modelling process of final Energy Consumption (*Consum d'energia final*) and with the same process, only report the results of CO2 emission prediction.

5-1 Consum d'energia final

5-1-1 Feature Selection

First of all, I extracted the most correlated features with *Consum d'energia final*. Figure 22 illustrates the correlation bar chart of this feature. As it can be seen, there are lots of features that are moderately to highly correlated with the target including age, heating efficiency, Cost, total energy, total emission and ... . I applied a filter of 0.15 of correlation to choose a smaller subset of features to work with. The filtering process gave me a list of 32 features (including the target itself). In the next step, I calculated the selected features pairwise correlation using the heatmap to remove high-correlated features. Figure 23 shows the result of pairwise correlation matrix. As the heatmap plot shows, there are multiple features which are highly correlated to each other and must be removed. Here is the list of these features:

```
data = data.drop(columns=[
                         'pp_total_Emissions',
                         "preprocessed_Consum d'energia final",
                         "ENCODED Qualificació de consum d'energia primaria no renovable" ,
                         'preprocessed_Energia primària no renovable',
                         "ENCODED Qualificacio d'emissions de CO2",
                         'ENCODED Qualificació emissions calefacció',
                         'ENCODED Qualificació emissions refrigeració',
                         'ENCODED Qualificació emissions ACS',
                         'ENCODED Qualificació energia refrigeració',
                         'ENCODED Qualificació energia ACS',
                         'ENCODED Qualificació energia calefacció demanda',
                         'ENCODED Qualificació energia refrigeració demanda',
                          'heating_efficiency',
                         'VEHICLE ELECTRIC_other',
                          'VEHICLE ELECTRIC_NO',
                         'XARXA DISTRICTE_NO',
                         'XARXA DISTRICTE_other',
                         'REHABILITACIO_ENERGETICA_NO',
                         'REHABILITACIO_ENERGETICA_other',
                          'Energia primària no renovable',
                 ])
```

Figure 22 Correlation bar chart of *Consum d'energia final*

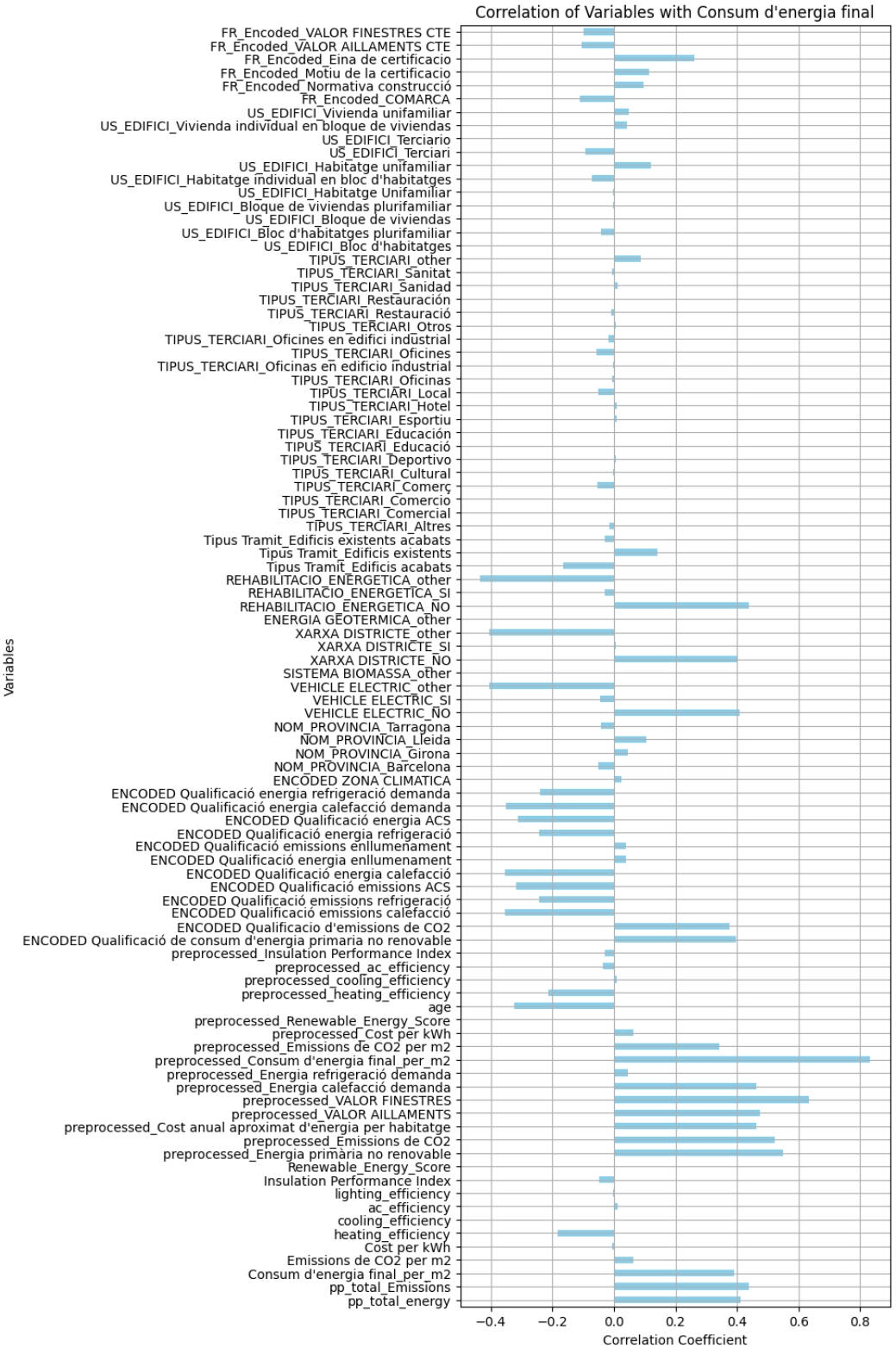
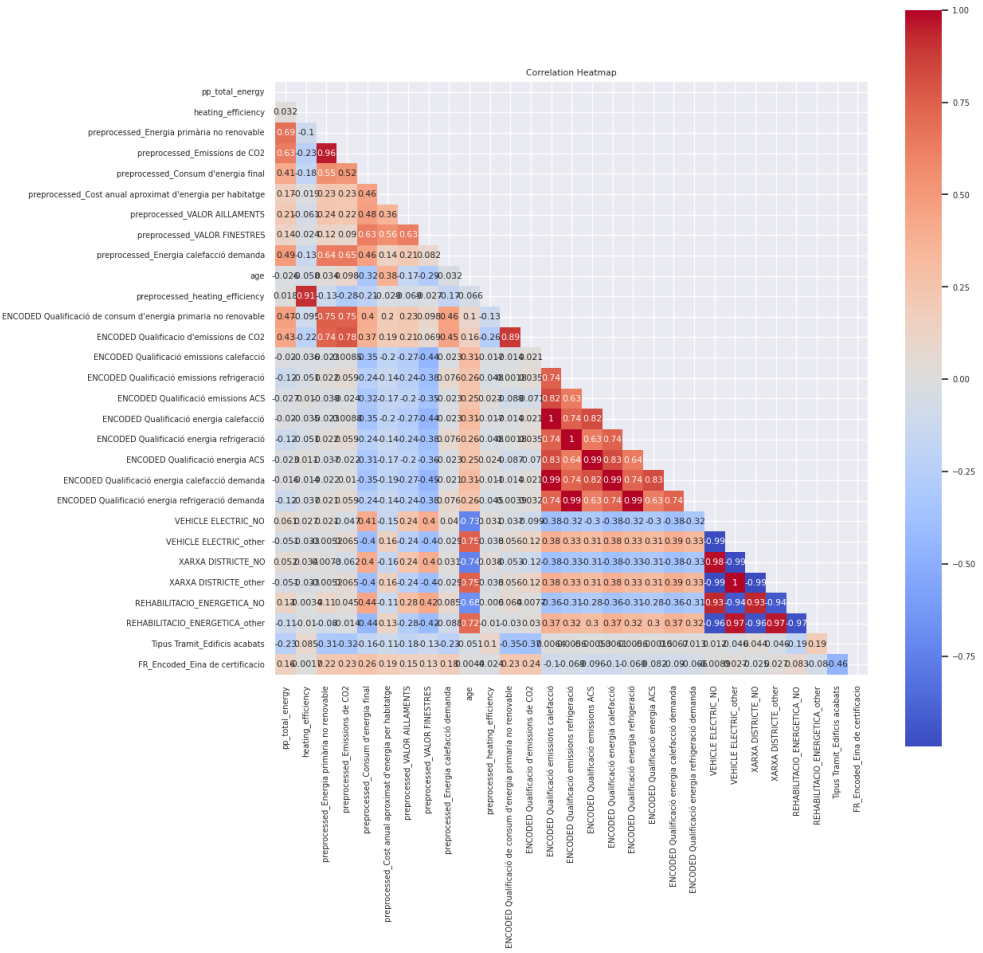
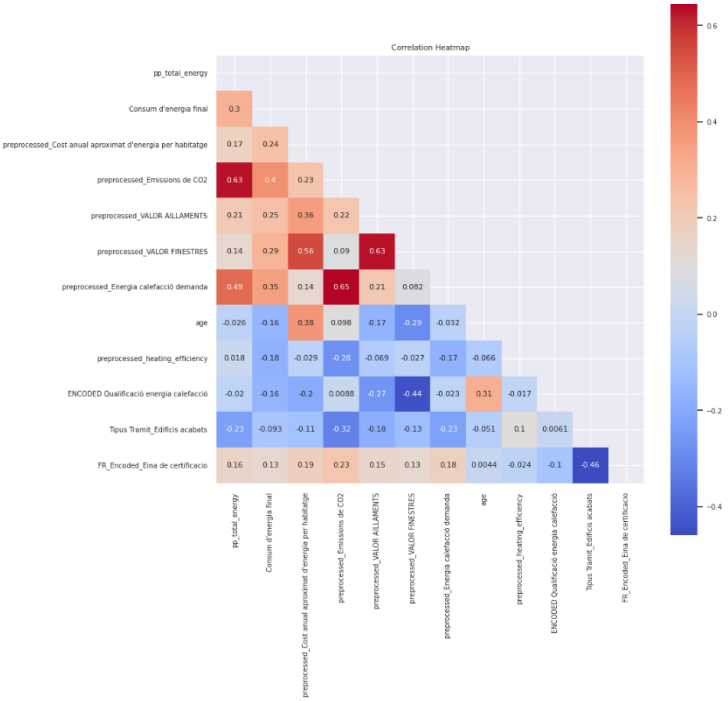


Figure 22 Heatmap Pairwise Correlation of selected features



After removing the highly-correlated features, we ended up with this correlation matrix which determines everything is ready for training a model. The red points show a correlation of 0.63-0.65 but I decided to ignore them and run the model to see the result.



5-1-2 Model Configuration

I applied a grid search for each of the ensemble models to find the best hyper parameters. Since the dataset was relatively large, grid searching took too much time especially for RF model. To cope with this issue, I used a small subset of dataset around 213,908 records to find the optimal hyper-parameters and then applied them to the model while training on real data. I split 0.2 of data as the test set with the size of 267,385 records and the rest of data as the train set. Three accuracy metric MSE, MAE and R^2 represents the performance of each model. Figure 23 shows the list of tuned parameters for these three models:

Figure 23 Tuned hyper-parameters for three **RF, XGB** and **Cat-boost models**

```
XGBRegressor(n_estimators=100,
             random_state=42,
             use_label_encoder=False,
             eval_metric='rmse',
             gamma = 1.0,
             learning_rate = 0.1,
             max_depth = 4,
             subsample = 0.5
             )
```

```
CatBoostRegressor(iterations=300,
                  random_seed=42,
                  verbose=0,
                  border_count = 64,
                  depth = 5,
                  l2_leaf_reg=0.5,
                  learning_rate=0.1
                  )
```

```
RandomForestRegressor(n_estimators=100,
                      random_state=42,
                      max_depth=None,
                      max_features= 0.75,
                      min_samples_leaf=1,
                      min_samples_split =4
                      )
```

5-1-3 Results

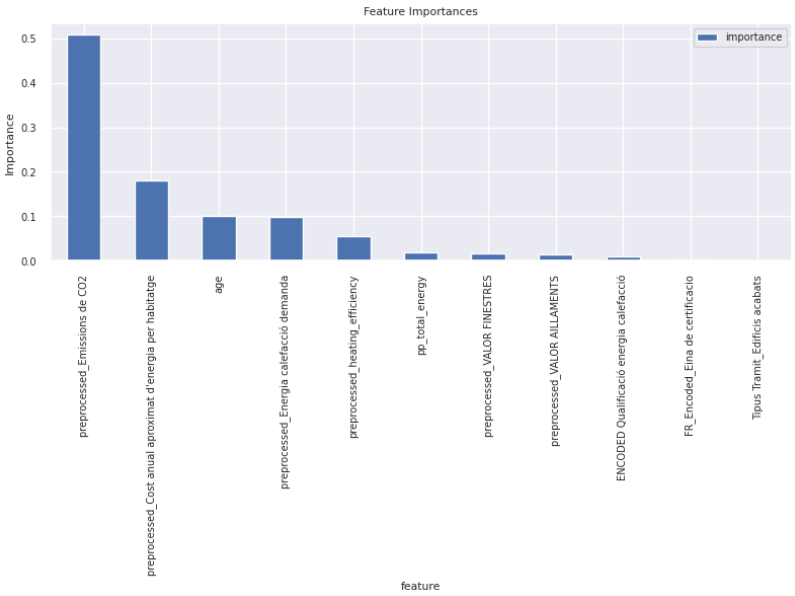
Random Forest: this mode performed relatively good on the data, but it took longer time to be executed.

- **Mean Squared Error**: 0.5702114530173301
- **Mean Absolute Error**: 0.04407511957616473
- **R^2 Score**: 0.21741453975808245

These performances, reveals some inconsistencies. The **Mean Squared Error (MSE)** is significantly higher at approximately 0.57, indicating larger squared errors and potentially more variance in the predictions. However, the **Mean Absolute Error (MAE)** is surprisingly low at about 0.044, suggesting that while the model may not be capturing the full range of variability, it is making fewer extreme errors. The **R^2 Score** of 0.217 indicates that only about 22% of the variance in the dependent variable is explained by the model, which is a relatively poor fit.

Figure 24 shows important features recognized by RF model. CO2 emission, annual cost and age are the top 3 important features for training.

Figure 24 feature importance of **RF**

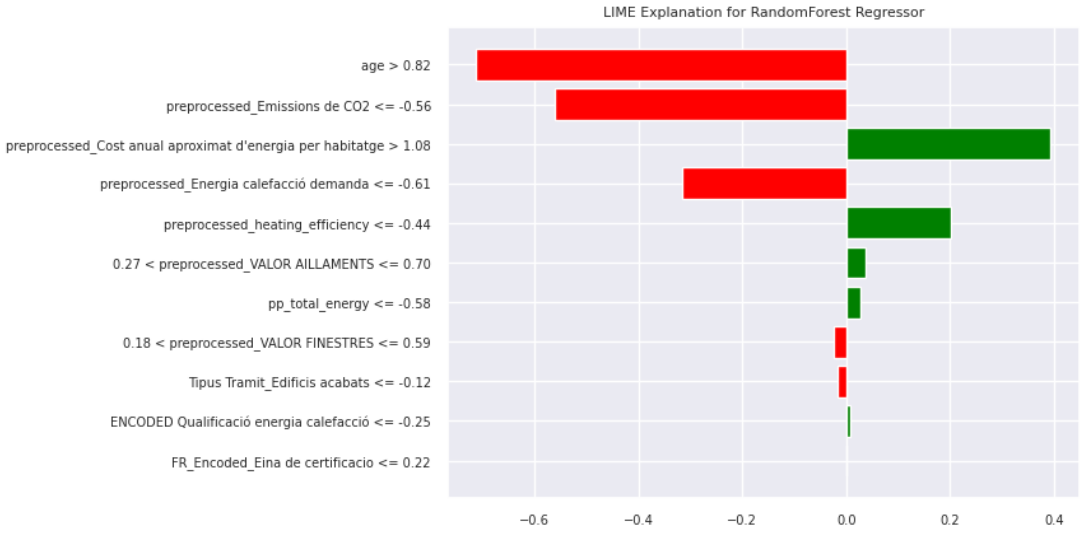


Feature Importances:

	feature	importance
2	preprocessed_Emissions de CO2	0.509025
1	preprocessed_Cost anual aproximat d'energia pe...	0.180319
6	age	0.100394
5	preprocessed_Energia calefacció demanda	0.097551
7	preprocessed_heating_efficiency	0.055123
0	pp_total_energy	0.017796
4	preprocessed_VALOR FINESTRES	0.016507
3	preprocessed_VALOR AILLAMENTS	0.012552
8	ENCODED Qualificació energia calefacció	0.008037
10	FR_Encoded_Eina de certificacio	0.002626
9	Tipus Tramit_Edificis acabats	0.000070

Figure 25 illustrate the interpretability of RF model using Lime for predicting the test sample with index **478,659.** The LIME plot indicates the local feature importance for a single prediction**.** The most significant factors pushing the prediction towards a lower value (shown in red) are 'age > 0.82', *'preprocessed_Emissions de CO2* <= -0.56', *'preprocessed_Energia calefacció demanda* <= -0.61', while *'preprocessed_Cost anual aproximat d'energia per habitatge* > 1.08' contributes the most to increasing the predicted value (shown in green). Other features have smaller positive or negative influences on the prediction.

Figure 25 interpretability of **RF** using Lime

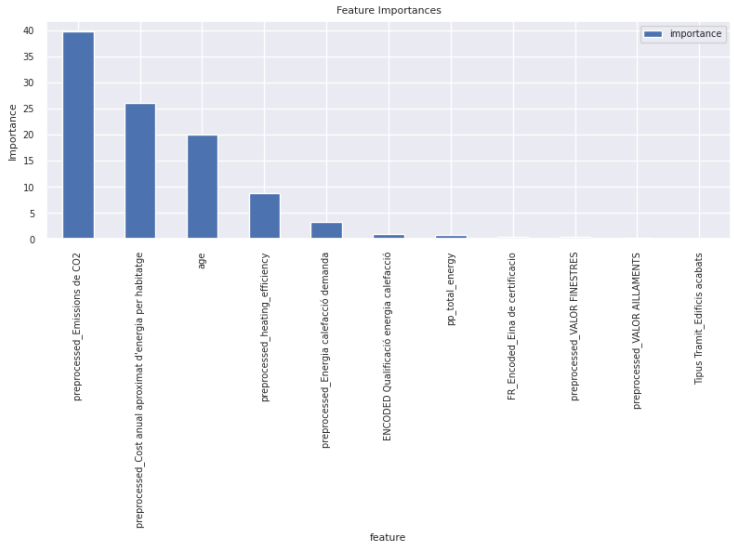


Catboost: this mode performed much better than RF and also much faster. Here are the accuracy results of Catboost:

- **Mean Squared Error:** 0.083907367536856
- **Mean Absolute Error:** 0.11042580215548493
- **R^2 Score:** 0.7984444379565124

This performance indicates a strong predictive capability of this model. The **Mean Squared Error (MSE)** of approximately 0.084 and the **Mean Absolute Error (MAE)** of about 0.110 suggest that the model is generally accurate, with a slight bias towards underestimation given the MAE is slightly higher than the square root of MSE. However, the **R^2 Score** of 0.798 indicates that nearly 80% of the variance in the dependent variable is explained by the model, which is a strong indicator of its effectiveness. Overall, these metrics suggest that the Catboost Regressor is well-suited for this regression task, providing reliable predictions with minimal error.

Figure 26 Catboost Important features



Feature Importances:

	feature	importance
2	preprocessed_Emissions de CO2	39.783738
1	preprocessed_Cost anual aproximat d'energia pe...	25.973569
6	age	19.931937
7	preprocessed_heating_efficiency	8.686319
5	preprocessed_Energia calefacció demanda	3.297041
8	ENCODED Qualificació energia calefacció	0.920079
0	pp_total_energy	0.640443
10	FR_Encoded_Eina de certificacio	0.308397
4	preprocessed_VALOR FINESTRES	0.300356
3	preprocessed_VALOR AILLAMENTS	0.156224
9	Tipus Tramit_Edificis acabats	0.001895

The feature importance list from CatBoost highlights that the same as RF model, *Emissions de CO2* and *Cost* and *Age* are the most influential features, contributing nearly 86% of the model's predictive power. **heating_efficiency** also plays a significant role, contributing about 9%. The remaining features have much less impact. This suggests that environmental and cost-related factors, along with age, are key predictors in the model.

Figure 26 interpretability of **Catboost** using Lime

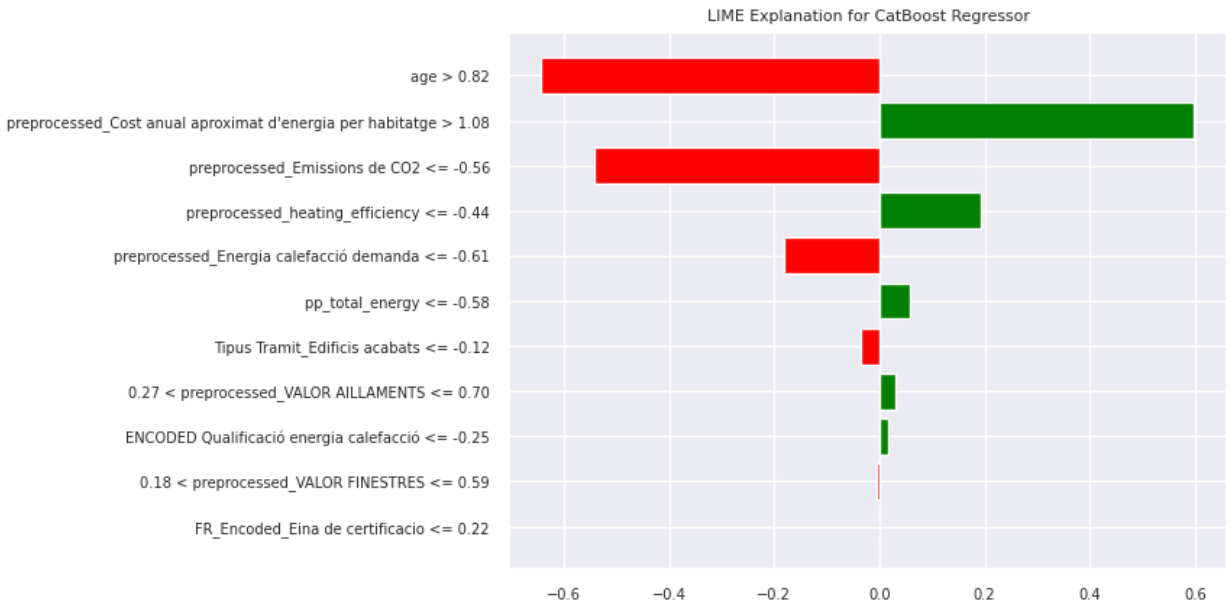


Figure 26 interpretability of **Catboost** using SHAP

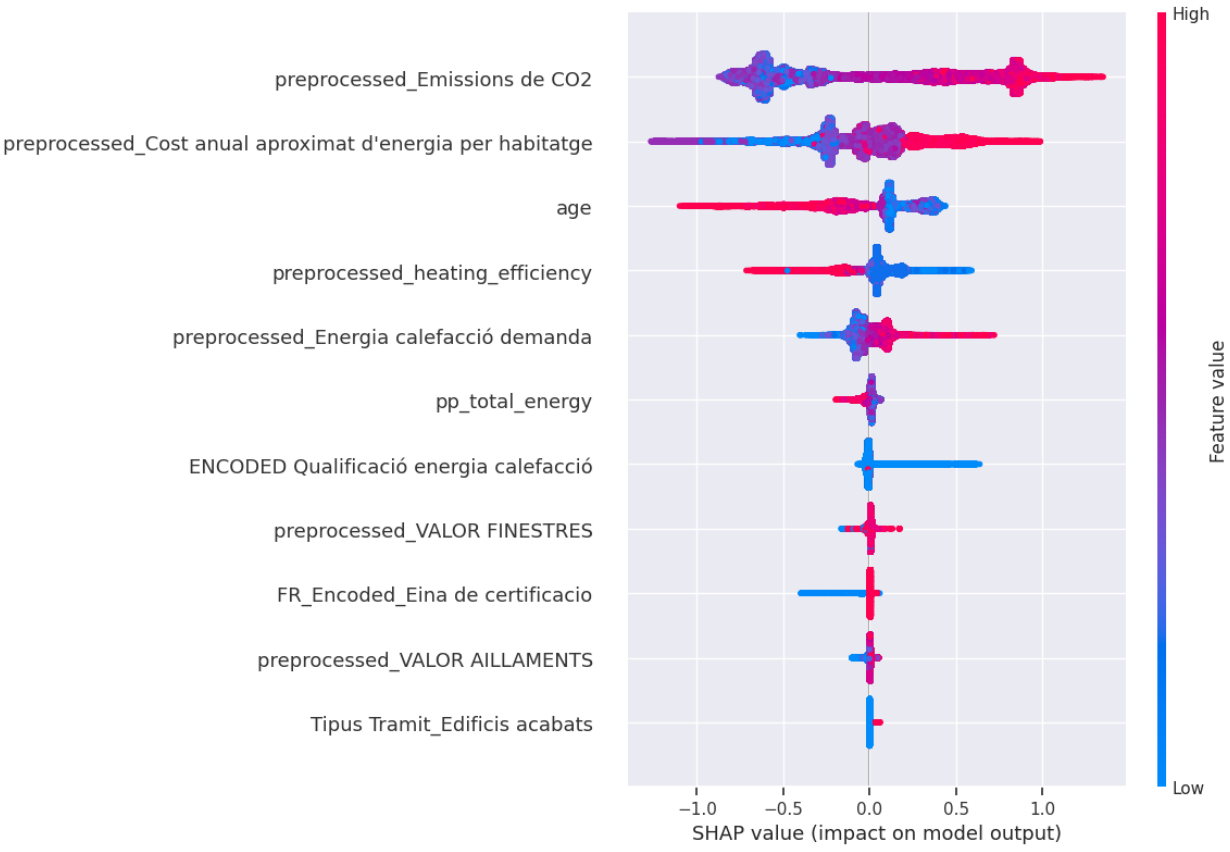


Figure 25 illustrate lime plot of performance of catboost pretrained model. LIME provides a snapshot of feature importance for a specific instance, in this case index of **478,659.** Based on the figure, the most important feature contributing positively to the prediction is "*preprocessed_Cost anual aproximat d'energia per habitatge* > 1.08 and *heating efficiency* <-0.44 ". The most important features contributing negatively are "*age* > 0.82" and "*emsission_de CO2* < - 0.56" .

The SHAP summary plot provides a global view of feature importance and their impact on the model output. Features are ranked by importance. For example, "preprocessed_Emissions de CO2" appears to be the most important feature. The color indicates the feature value (red = high, blue = low), and the x-axis shows the SHAP value (impact on model output). We can see, for instance, that a high value of "preprocessed_Emissions de CO2" tends to decrease the model output, while a high value of "preprocessed_Cost anual aproximat d'energia per habitatge" tends to increase the model output. SHAP values provide a more comprehensive understanding of feature effects across the entire dataset.

XGBoost: Here are the accuracy results of xgboost model.

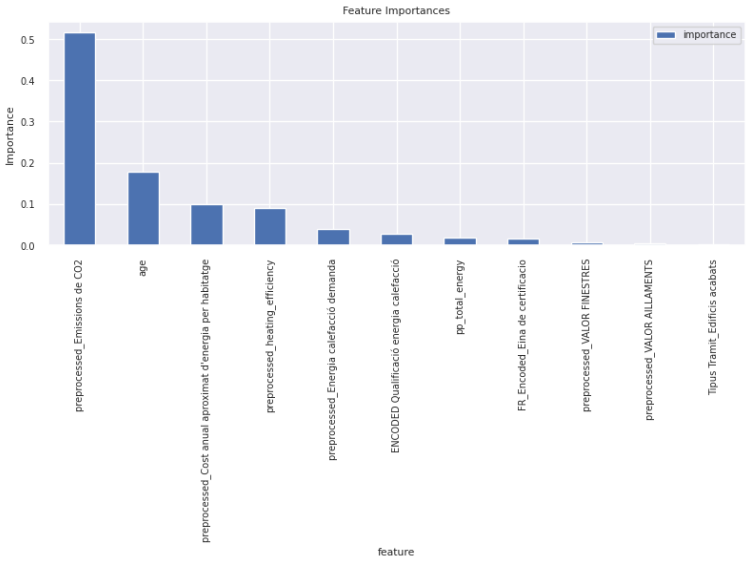
- **Mean Squared Error:** 0.0734779691576633
- **Mean Absolute Error:** 0.12646084136350222
- **R^2 Score:** 0.8182046902330615

The XGBoost model regressor shows strong performance with a Mean Squared Error (MSE) of 0.0735 and a Mean Absolute Error (MAE) of 0.1265. The R^2 Score of 0.8182 indicates that the model explains about 82% of the variance in the target variable, suggesting a good fit. Compared to the Random Forest (RF) model, XGBoost performs significantly better in terms of MSE and R^2 Score, although RF has a slightly lower MAE. However, the overall performance of XGBoost is superior due to its higher R^2 Score and lower MSE.

In comparison with CatBoost, XGBoost has a slightly lower MSE and a higher R^2 Score, indicating a marginally better performance. CatBoost's MAE is slightly lower than XGBoost's, but the difference is not substantial. Overall, XGBoost appears to be the most accurate model among the three, offering a good balance between error metrics and explanatory power. Its ability to explain a larger portion of the variance in the data makes it a preferable choice for this particular regression task.

Figure 27 illustrates important features recognized by XGBoost. The feature importance from XGBoost highlights that , *Emissions de CO2* and *Age* are the most significant influential features, contributing nearly 69% of the model's predictive power. The remaining features have much less impact. This suggests that the capability of this model highly depends on these two features.

Figure 27 XGBoost Important features



Feature Importances:

	feature	importance
2	preprocessed_Emissions de CO2	0.516245
6	age	0.178885
1	preprocessed_Cost anual aproximat d'energia pe...	0.099236
7	preprocessed_heating_efficiency	0.089184
5	preprocessed_Energia calefacció demanda	0.038283
8	ENCODED Qualificació energia calefacció	0.028796
0	pp_total_energy	0.019156
10	FR_Encoded_Eina de certificacio	0.015585
4	preprocessed_VALOR FINESTRES	0.007833
3	preprocessed_VALOR AILLAMENTS	0.004944
9	Tipus Tramit_Edificis acabats	0.001852

Figure 28 and 29 provide SHAP and Lime interpretation for XGboost model. The SHAP summary plot ranks features by importance and shows the distribution of their impact on the model output. "*preprocessed_Emissions de CO2*" appears to be the most important feature. Higher values of this feature (indicated by red) tend to decrease the model output (negative SHAP values), while lower values (blue) increase it. Other important features include "*preprocessed_Cost anual aproximat d'energia per habitatge*" and "*age*," both showing a similar pattern where higher values tend to decrease the model output and vice versa. The spread of points for each feature indicates the variability of their impact across different samples.

The LIME plot provides a local explanation for individual predictions. The features are ranked by their contribution to a specific prediction. In this particular instance, higher values of "*preprocessed_Cost anual aproximat d'energia per habitatge*" contribute positively (green) to the prediction, while higher values of "age" and "*preprocessed_Emissions de CO2*" contribute negatively (red). The numbers indicate the feature values and their corresponding impact on the model output for a particular instance. For example, when "*preprocessed_Emissions de CO2*" is less than or equal to -0.56, it negatively affects the predicted value.

Figure 28 interpretability of **XGboost** using Lime

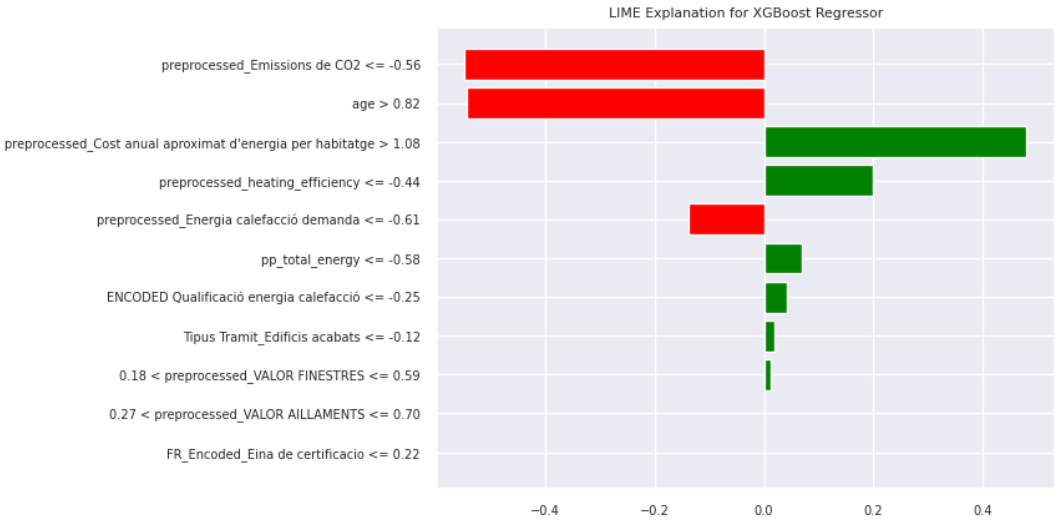
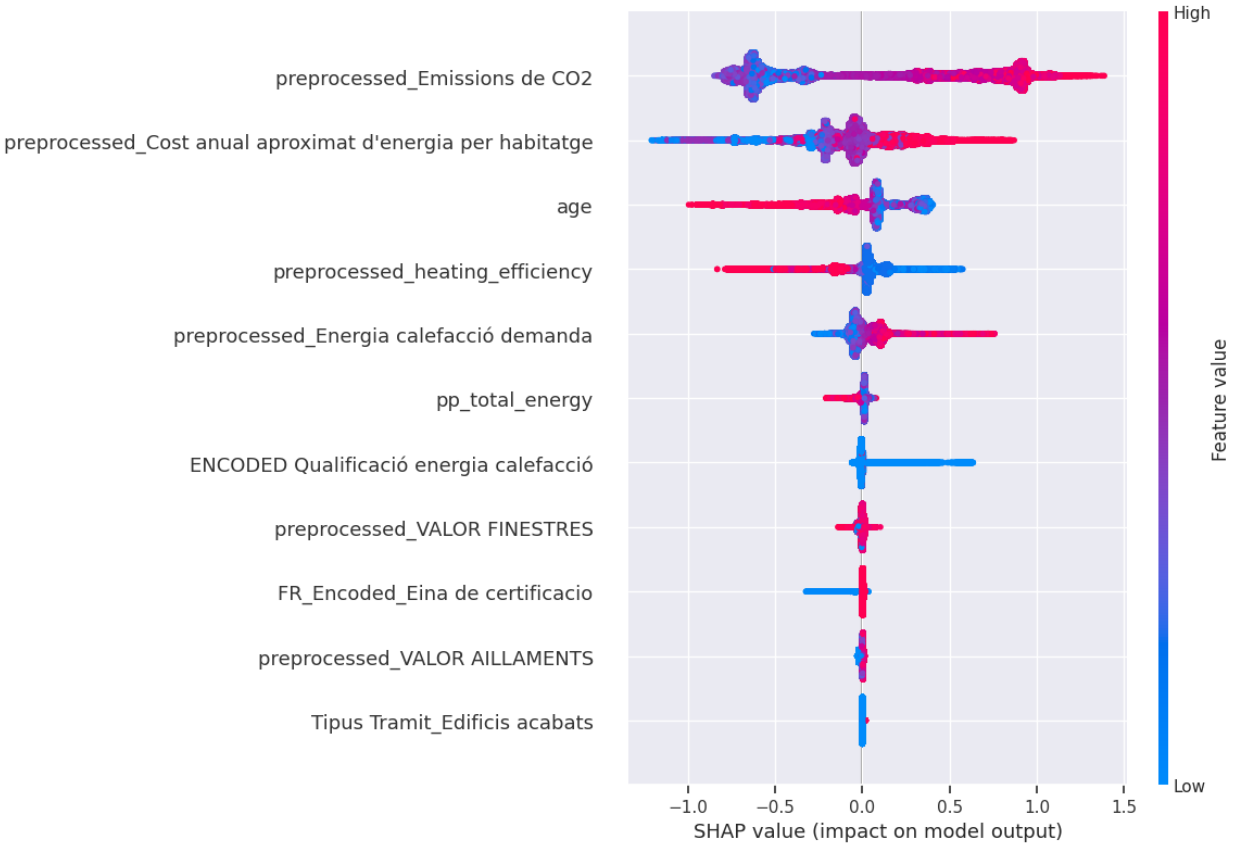


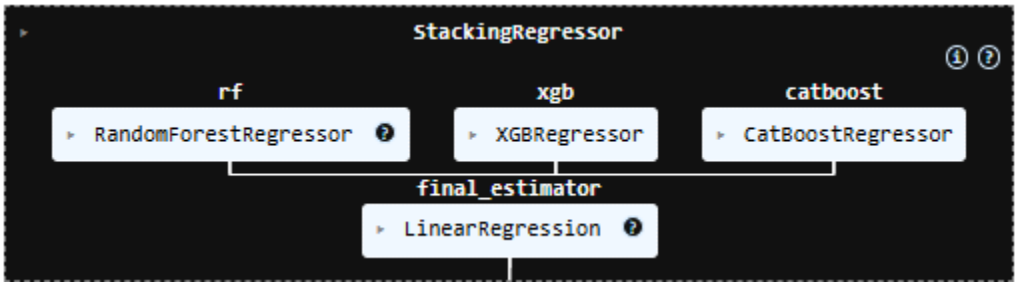
Figure 29 interpretability of **XGboost** using SHAP



Stacking Regressor Model for Aggregation

After training all three ensemble models, I passed them through a simple linear regression model to have a majority vote of their results. Stacking the models also improve the robustness of models and is a good way to boost the performance. Figure 30 shows a plain structure of the stacking model.

Figure 30 stacking the **rf**, **xgb** and **catbbost** models



Here is the accuracy of the final estimator on the same set of train and test:

- Mean Squared Error: 0.04083343863681367
- Mean Absolute Error: 0.06565375904333184
- R^2 Score: 0.8976289933862698

Looking at the results, we can see that the Stacking Regressor demonstrates impressive performance. It achieves a Mean Squared Error (MSE) of 0.0408 and a Mean Absolute Error (MAE) of 0.0657, both of which are significantly lower than those of the individual models. The R^2 Score of 0.8976 indicates that the Stacking Regressor explains nearly 90% of the variance in the target variable, showing a substantial improvement over the individual models.

Compared to XGBoost, CatBoost, and Random Forest, the Stacking Regressor outperforms all three in terms of MSE, MAE, and R^2 Score. Specifically, it reduces the MSE by about 45% compared to XGBoost and improves the R^2 Score by about 8%. This suggests that combining the strengths of multiple models through stacking can lead to more accurate predictions than relying on any single model alone. The Stacking Regressor's ability to leverage the diverse predictions of XGBoost, CatBoost, and Random Forest results in a more robust and accurate regression model.

5-2 Emission de CO2

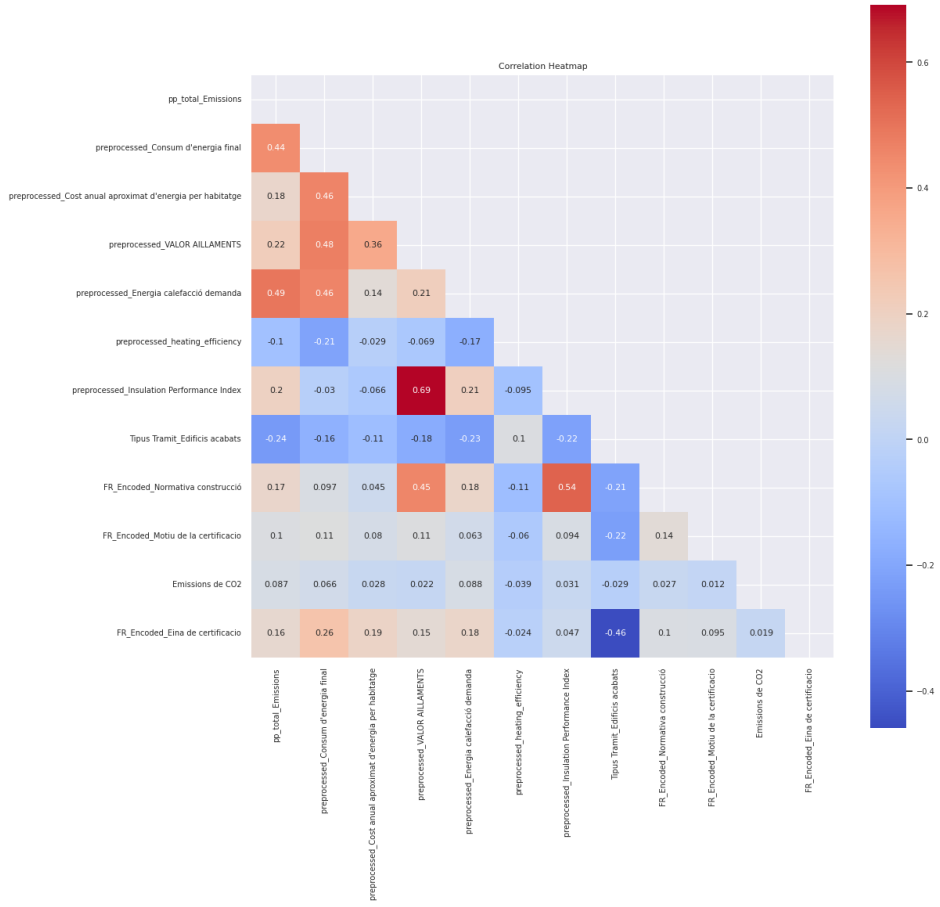
The Catalunya dataset includes consumption and emission data for two primary energy sources and CO2. Therefore, I selected these two key targets for modeling to examine their outcomes and facilitate comparison. As the modeling workflow for both targets is identical, this section will concentrate on the most significant results and analyses to maintain a concise and focused report.

Figure 31 presents a bar chart correlation matrix for **Emission de CO2**, revealing that numerous features exhibit moderate to high correlations with **Emission de CO2**. To refine the feature selection, I applied a correlation threshold of 0.15 and removed highly pairwise correlated features. The resulting final set of selected features is illustrated in Figure 32.

Figure 31


Correlation of Variables with Emissions de CO2
Variables
FR_Encoded_VALOR FINESTRES CTE
FR_Encoded_VALOR AILLAMENTS CTE
FR_Encoded_Eina de certificacio
FR_Encoded_Motiu de la certificacio
FR_Encoded_Normativa construcció
FR_Encoded_COMARCA
US_EDIFICI_Vivienda unifamiliar
US_EDIFICI_Vivienda individual en bloque de viviendas
US_EDIFICI_Terciario
US_EDIFICI_Terciari
US_EDIFICI_Habitatge unifamiliar
US_EDIFICI_Habitatge individual en bloc d'habitatges
US_EDIFICI_Habitatge Unifamiliar
US_EDIFICI_Bloque de viviendas plurifamiliar
US_EDIFICI_Bloque de viviendas
US_EDIFICI_Bloc d'habitatges plurifamiliar
US_EDIFICI_Bloc d'habitatges
TIPUS_TERCIARI_other
TIPUS_TERCIARI_Sanitat
TIPUS_TERCIARI_Sanidad
TIPUS_TERCIARI_Restauración
TIPUS_TERCIARI_Restauració
TIPUS_TERCIARI_Otros
TIPUS_TERCIARI_Oficines en edifici industrial
TIPUS_TERCIARI_Oficines
TIPUS_TERCIARI_Oficinas en edificio industrial
TIPUS_TERCIARI_Oficinas
TIPUS_TERCIARI_Local
TIPUS_TERCIARI_Hotel
TIPUS_TERCIARI_Esportiu
TIPUS_TERCIARI_Educación
TIPUS_TERCIARI_Educació
TIPUS_TERCIARI_Deportivo
TIPUS_TERCIARI_Cultural
TIPUS_TERCIARI_Comerç
TIPUS_TERCIARI_Comercio
TIPUS_TERCIARI_Comercial
TIPUS_TERCIARI_Altres
Tipus Tramit_Edificis existents acabats
Tipus Tramit_Edificis existents
Tipus Tramit_Edificis acabats
REHABILITACIO_ENERGETICA_other
REHABILITACIO_ENERGETICA_SI
REHABILITACIO_ENERGETICA_NO
ENERGIA GEOTERMICA_other
XARXA DISTRICTE_other
XARXA DISTRICTE_SI
XARXA DISTRICTE_NO
SISTEMA BIOMASSA_other
VEHICLE ELECTRIC_other
VEHICLE ELECTRIC_SI
VEHICLE ELECTRIC_NO
NOM_PROVINCIA_Tarragona
NOM_PROVINCIA_Lleida
NOM_PROVINCIA_Girona
NOM_PROVINCIA_Barcelona
ENCODED ZONA CLIMATICA
ENCODED Qualificació energia refrigeració demanda
ENCODED Qualificació energia calefacció demanda
ENCODED Qualificació energia ACS
ENCODED Qualificació energia refrigeració
ENCODED Qualificació emissions enllumenament
ENCODED Qualificació energia enllumenament
ENCODED Qualificació energia calefacció
ENCODED Qualificació emissions ACS
ENCODED Qualificació emissions refrigeració
ENCODED Qualificació emissions calefacció
ENCODED Qualificacio d'emissions de CO2
ENCODED Qualificació de consum d'energia primaria no renovable
preprocessed_Insulation Performance Index
preprocessed_ac_efficiency
preprocessed_cooling_efficiency
preprocessed_heating_efficiency
age
preprocessed_Renewable_Energy_Score
preprocessed_Cost per kWh
preprocessed_Emissions de CO2 per m2
preprocessed_Consum d'energia final_per_m2
preprocessed_Energia refrigeració demanda
preprocessed_Energia calefacció demanda
preprocessed_VALOR FINESTRES
preprocessed_VALOR AILLAMENTS
preprocessed_Cost anual aproximat d'energia per habitatge
preprocessed_Consum d'energia final
preprocessed_Energia primària no renovable
Renewable_Energy_Score
Insulation Performance Index
lighting_efficiency
ac_efficiency
cooling_efficiency
heating_efficiency
Cost per kWh
Emissions de CO2 per m2
Consum d'energia final_per_m2
pp_total_Emissions
pp_total_energy
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Correlation Coefficient

Figure 32 Correlation Matrix of Final selected features



Models Hyper Parameters:

After applying the grid search on a subset of data the following parameters were obtained as the optimum params of each model.

```
XGBRegressor(n_estimators=100,
             random_state=42,
             use_label_encoder=False,
             eval_metric='rmse',
             gamma = 0,
             learning_rate = 0.1,
             max_depth = 4,
             subsample = 1.0
             )
```

```
CatBoostRegressor(iterations=300,
                  random_seed=42,
                  verbose=0,
                  border_count = 64,
                  depth = 5,
                  l2_leaf_reg=0,
                  learning_rate=0.1
                  )
```

```
RandomForestRegressor(n_estimators=100,
                      random_state=42,
                      max_depth=None,
                      max_features= 0.5,
                      min_samples_leaf=3,
                      min_samples_split =2
                      )
```

Results:

	MSE	MAE	R^2
Random Forest	0.12089666632600303	0.13737208872732026	0.7326848398995576
Cat boost	0.8339671828430775	0.026368806928338904	0.005326220137494286
XGboost	0.19522896904889783	0.2029360631550762	0.5760531507314118
Stacking regressor	0.1195328046499827	0.13751483559909694	0.7357004805566549

Looking at the results, the Stacking Regressor and Random Forest models demonstrate the best performance overall. They both have very similar, low MSE (Mean Squared Error) and MAE (Mean Absolute Error) values, indicating accurate predictions with minimal error. Furthermore, their high R2 values (0.7357 and 0.7327, respectively) suggest they explain a large proportion of the variance in the data.

In contrast, CatBoost performs poorly, with a high MSE and a near-zero R2 score, suggesting it is not a good fit for this dataset. XGBoost falls in the middle, with an MSE and MAE higher than Random Forest and Stacking Regressor, and an R2 value that indicates a moderate level of explained variance. The Stacking Regressor narrowly edges out Random Forest as the best-performing model, though the difference is marginal.

Figure 33 Feature Importance of **RF**

Feature Importances:

	feature	importance
1	preprocessed_Consum d'energia final	0.469803
4	preprocessed_Energia calefacció demanda	0.174281
0	pp_total_Emissions	0.123845
2	preprocessed_Cost anual aproximat d'energia pe...	0.110220
5	preprocessed_heating_efficiency	0.067944
6	preprocessed_Insulation Performance Index	0.020284
3	preprocessed_VALOR AILLAMENTS	0.017109
9	FR_Encoded_Motiu de la certificacio	0.007806
8	FR_Encoded_Normativa construcció	0.006421
10	FR_Encoded_Eina de certificacio	0.002080
7	Tipus Tramit_Edificis acabats	0.000208

Figure 34 Feature Importance of **Catboost**

Feature Importances:

	feature	importance
1	preprocessed_Consum d'energia final	45.796895
2	preprocessed_Cost anual aproximat d'energia pe...	13.101717
4	preprocessed_Energia calefacció demanda	13.100444
0	pp_total_Emissions	11.699416
5	preprocessed_heating_efficiency	9.804266
3	preprocessed_VALOR AILLAMENTS	2.318833
6	preprocessed_Insulation Performance Index	1.796363
9	FR_Encoded_Motiu de la certificacio	1.091679
8	FR_Encoded_Normativa construcció	0.649480
10	FR_Encoded_Eina de certificacio	0.595096
7	Tipus Tramit_Edificis acabats	0.045812

Figure 35 Feature Importance of **xgboost**

```
Feature Importances:
                                        feature  importance
1                preprocessed_Consum d'energia final    0.485409
4             preprocessed_Energia calefacció demanda    0.128358
3                       preprocessed_VALOR AILLAMENTS    0.120179
0                                  pp_total_Emissions    0.108100
2   preprocessed_Cost anual aproximat d'energia pe...    0.049927
5                    preprocessed_heating_efficiency    0.035660
6           preprocessed_Insulation Performance Index    0.032125
8                     FR_Encoded_Normativa construcció    0.016184
10                    FR_Encoded_Eina de certificacio    0.009633
9                 FR_Encoded_Motiu de la certificacio    0.009622
7                       Tipus Tramit_Edificis acabats    0.004803
```

The feature importance plots reveal that "preprocessed_Consum d'energia final" is consistently the most important feature across all three models (Random Forest, XGBoost, and CatBoost). Other significant features include "preprocessed_Energia calefacció demanda", "pp_total_Emissions", and "preprocessed_Cost anual aproximat d'energia per...", although their relative importance varies slightly between the models. Features like "FR_Encoded_Motiv de la certificacio", "FR_Encoded_Normativa construcció", and "Tipus Tramit_Edificis acabats" consistently rank among the least important features, suggesting they have minimal impact on the models' predictive capabilities.

Bonus Question

I found it hard to look for a real data with the exact features that I needed for my model, however I considered a synthetic sample with some similar charecteristics of Persian houses (As I live in this region) and predicted it.

Here are the available features of this sample :

- Tipus Tramit = 'Edificis acabats' (finished building)
- emission de co2 = 35
- DATA_ENTRADA : 29/01/2022 === > age= 3
- cooling energy : 16 kw/m2
- heating energy : 54 kw/m2
- heating emission : 10 kw/m2
- Qualificació energia calefacció (heating rate) : D
- VALOR AILLAMENTS (Medium façade transmittance) : 0.73 W / m2 · K
- VALOR FINESTRES (Average window transmittance : 4 W / m2 · K

I filled the other values with median of the corresponding columns. Figure 35 shows the sample before and after preprocessing.

Figure 36 sample data

Cost anual aproximat d'energia per habitatge	16.75
Emissions de CO2	34.96
VALOR AILLAMENTS	0.73
VALOR FINESTRES	4
Energia calefacció demanda	36.48
Qualificació energia calefacció	D
Tipus Tramit	Edificis acabats
Eina de certificacio	CE3X
DATA_ENTRADA	29/01/2022
Energia calefacció	54
Energia refrigeració	16
Energia ACS	137.84
Energia enllumenament	0.0
Emissions calefacció	10
Emissions calefacció	10
Emissions de CO2\t	35
age	3
DATA_ENTRADA\t	29/01/2022
VALOR AILLAMENTS\t	0.8

pp_total_energy	-0.08311
preprocessed_Cost anual aproximat d'energia per habitatge	-0.22563
preprocessed_Emissions de CO2	0.215355
preprocessed_VALOR AILLAMENTS	0.273779
preprocessed_VALOR FINESTRES	0.178779
preprocessed_Energia calefacció demanda	-1.547148
age	0.817
preprocessed_heating_efficiency	1.846062
ENCODED Qualificació energia calefacció	-0.249285
Tipus Tramit_Edificis acabats	0
FR_Encoded_Eina de certificacio	0.216439

After train the xgboost , rf and catboost model on this sample here is the result :

Figure 37 Rf output

```
y_pred =rf_model.predict(np.array(sample[features]).reshape(1, -1))
y_pred_boxcox = scaler.inverse_transform(y_pred.reshape(1, -1))
y_pred_original = inv_boxcox(y_pred_boxcox, boxcox_lambda) - 1
y_pred_original
```

```
array([[0.0813596]])
```

Figure 38 Xgboost output

```
y_pred =xgb_model.predict(np.array(sample[features]).reshape(1, -1))
y_pred_boxcox = scaler.inverse_transform(y_pred.reshape(1, -1))
y_pred_original = inv_boxcox(y_pred_boxcox, boxcox_lambda) - 1
y_pred_original
```

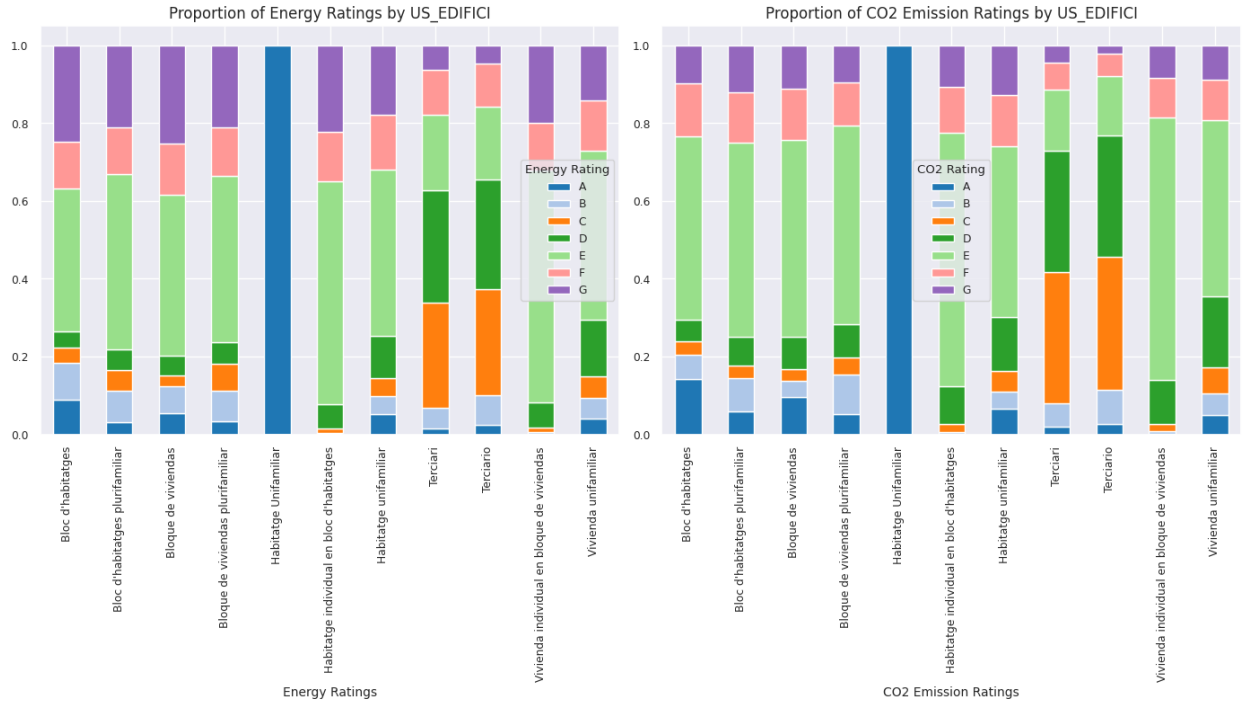
```
array([[-0.03504699]], dtype=float32)
```

Figure 39 Catboost output

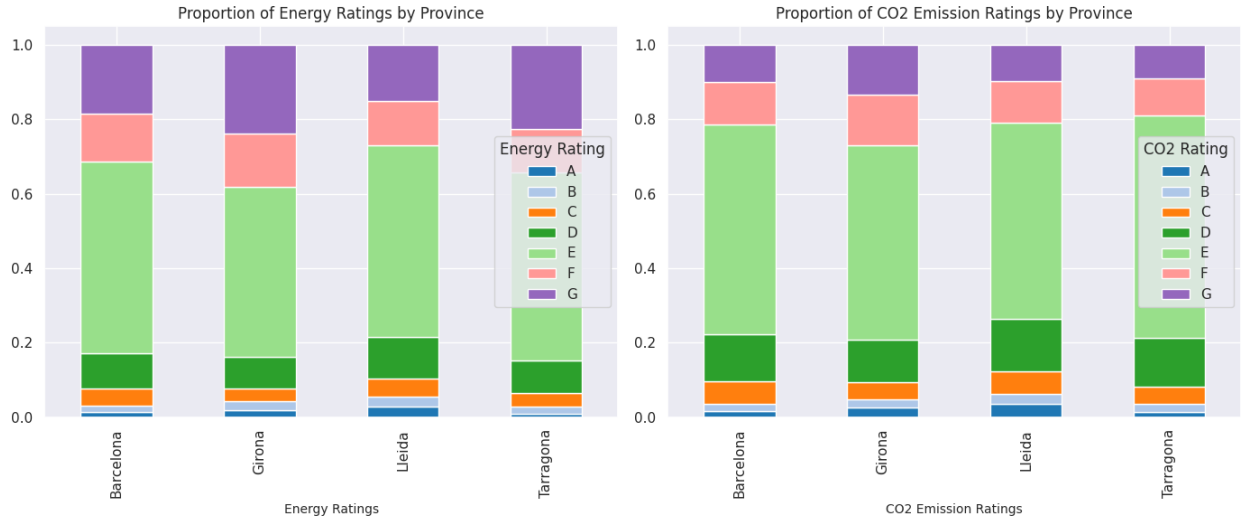
```
y_pred =catboost_model.predict(np.array(sample[features]).reshape(1, -1))
y_pred_boxcox = scaler.inverse_transform(y_pred.reshape(1, -1))
y_pred_original = inv_boxcox(y_pred_boxcox, boxcox_lambda) - 1
y_pred_original
```

```
array([[-0.09329844]])
```

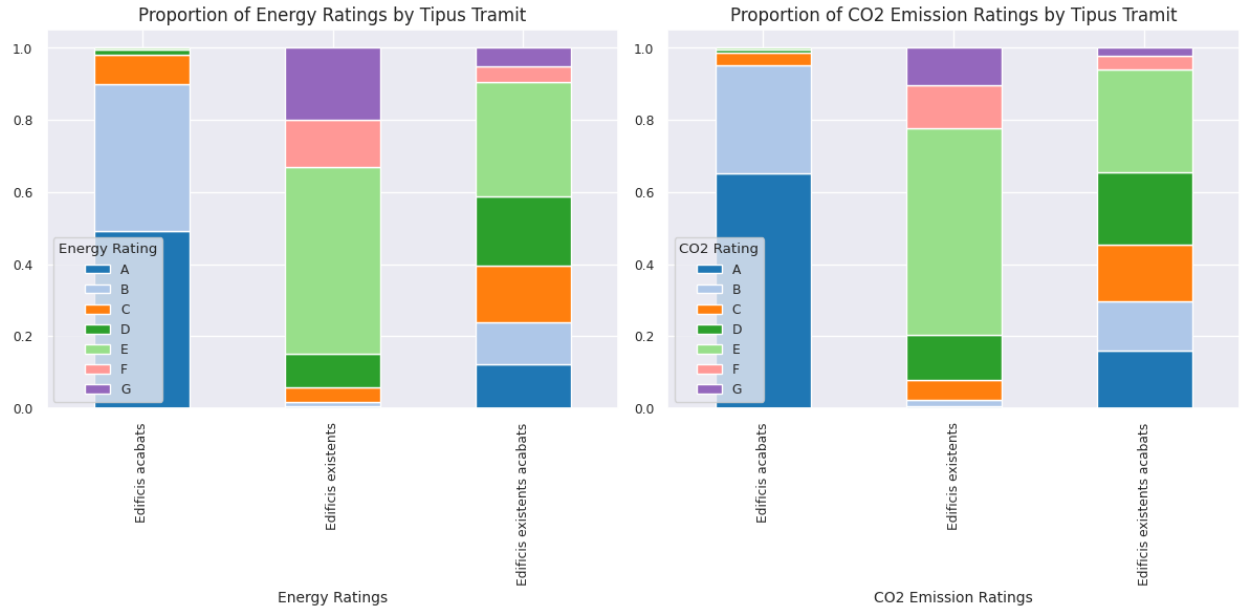
6- Appendix :



The image presents the proportion of energy and CO2 emission ratings for different types of buildings in Catalunya, ranging from A (most efficient) to G (least efficient). Single-family homes ('Habitatge Unifamiliar' / 'Vivienda unifamiliar') show a high proportion of A-rated energy consumption and CO2 emissions, in contrast to the multi-family apartment blocks ('Bloc d'habitatges plurifamiliar' / 'Bloque de viviendas plurifamiliar') which predominantly have E, F, and G ratings. Tertiary sector buildings ('Terciari' / 'Terciario') display a wide distribution of ratings, with a significant portion in the C, D, and E categories for both energy and CO2 emissions. A significant portion of individual housing in a block of apartments ("Habitatge individual en bloc d'habitatges"/"Vivienda individual en bloque de viviendas") consume less energy and produce fewer emissions.



The provided image shows the proportion of energy and CO2 emission ratings across the provinces of Catalunya: Barcelona, Girona, Lleida, and Tarragona. In both graphs, the ratings range from A to G, with A being the most efficient and G being the least. Generally, a significant portion of the ratings falls into the E category, with a notable amount also in the F and G categories, indicating a considerable opportunity for improvement in both energy efficiency and CO2 emissions across all provinces. Lleida and Tarragona have relatively similar distributions, while Barcelona and Girona also share similarities with each other, although there are still provincial differences to note when planning region-specific interventions.



The image presents two stacked bar charts comparing the proportion of energy ratings and CO2 emission ratings based on the "Tipus Tramit," which means "Type of Procedure". The x-axis labels "Edificis acabats," "Edificis existents," and "Edificis existents acabats" can be understood as "New Construction," "Existing Buildings," and "Existing Buildings after Refurbishment" respectively. For

energy ratings, new constructions predominantly fall into the 'A' and 'B' categories, indicating high energy efficiency. Existing buildings, on the other hand, are largely distributed across lower ratings like 'E', 'F' and 'G', implying lower energy efficiency. Refurbished existing buildings show an improvement, with a significant proportion moving towards 'C', 'D', and 'E' ratings. A similar trend is observed for CO2 emission ratings, where new constructions have a higher proportion of 'A' and 'B' ratings, existing buildings have poorer ratings, and refurbished buildings demonstrate an improvement in CO2 emission performance.